

CHAPTER 2

BRIEF REVIEW OF "ANALYSIS I" SEMINAR TOPICS

2.0 INTRODUCTION

Included in this section will be a brief review of some of the topics which were covered in the "ANALYSIS I" seminar. This particular chapter will review these particular topics:

2.1 WHAT IS VIBRATION AND HOW CAN IT BE USED TO EVALUATE MACHINERY CONDITION?

2.2 OVERVIEW OF THE STRENGTHS AND WEAKNESSES OF TYPICAL VIBRATION INSTRUMENTS

2.3 OVERVIEW OF VIBRATION TRANSDUCERS AND HOW TO PROPERLY SELECT THEM

2.4 UNDERSTANDING VIBRATION PHASE AND ITS APPLICATIONS

In addition to these topics, other items which were covered in "ANALYSIS I" will be briefly reviewed in other sections of the "ANALYSIS II" seminar text. However, the expressed purpose of Chapter 2 is to ensure everyone reviews the fundamentals before proceeding to more advanced topics.

2.1 WHAT IS VIBRATION AND HOW CAN IT BE USED TO EVALUATE MACHINERY CONDITION?

2.11 INTRODUCTION

Vibration is the response of a system to an internal or external stimulus causing it to oscillate or pulsate. While it is commonly thought that vibration itself damages machines and structures, it does not. Instead, *the damage is done by dynamic stress which causes fatigue of the materials;* and the *dynamic stresses are induced by vibration.* Equation 1 shows that the Vibration Amplitude is directly proportional to the Dynamic Force, and inversely proportional to the Dynamic Resistance in a spring-mass system like that shown in Figure 1. That is, *if two machines are subject to the same dynamic force, the amplitude response from the machine which has greater dynamic resistance will be less* than that of the other machine. For example, if a machine is placed on spring isolators, the vibration will likely increase due to less dynamic resistance for the same imposed dynamic forces. The transmission of vibration to the floor and surrounding structures will be less, but the vibration within the machine will likely increase. Yet, no additional damage will be done to the machine since the same *forces* (and therefore, fatigue stresses) will remain the same within this machine (as compared to when the machine was directly mounted to

$$\text{VIBRATION AMPLITUDE RESPONSE} \propto \frac{\text{DYNAMIC FORCE}}{\text{DYNAMIC RESISTANCE}} \quad \text{Eqn. 1}$$

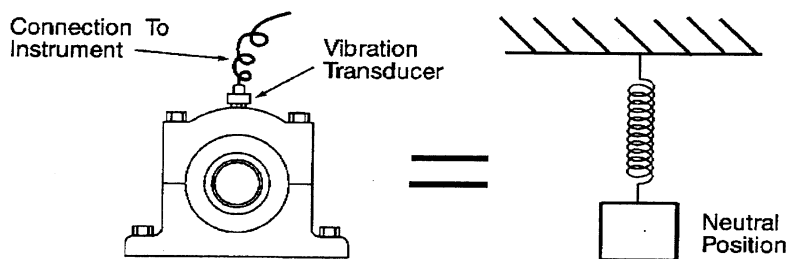


FIGURE 1
MASS IN NEUTRAL POSITION WITH NO APPLIED FORCE

the floor). Dynamic Resistance within a machine or structure is proportional to the amount of stiffness, damping and mass within the system. This will be discussed later in Chapter 11 which examines how these 3 parameters interact with one another.

Vibration has three important parameters which can be measured:

1. **Frequency** - How many times does the machine or structure vibrate per minute or per second?
2. **Amplitude** - How much vibration in mils, in/sec or g's?
3. **Phase** - How is the member vibrating in relation with a reference point?

2.12 WHAT IS VIBRATION FREQUENCY AND HOW DOES IT RELATE TO A TIME WAVEFORM?

Recall from an example of a pencil trace drawn on a strip chart recorder (if the pencil was fastened to a suspended mass which oscillates up and down on a spring), a uniform series of sine waves would be drawn. Each sine wave would represent one completed cycle - the mass would go from its neutral position to an upper limit of travel, down through its neutral position, then down to a lower limit of travel, and finally back to its neutral position (this completes one cycle of motion). Figure 2 shows how frequency can be calculated from it by measuring the time period (T) of one cycle (sec/cycle) and inverting to determine the frequency (cycles/sec). This is an example of a time waveform which plots Vibration Amplitude versus Time. This waveform is a truly sinusoidal waveform from which direct comparisons can be made between its Peak-to-Peak, Peak and RMS amplitudes (this will be covered in another section).

Frequency is expressed in either Cycles per Minute (CPM) or in Cycles per Second (CPS), which is now called Hertz (where 1 Hertz or Hz = 60 CPM).

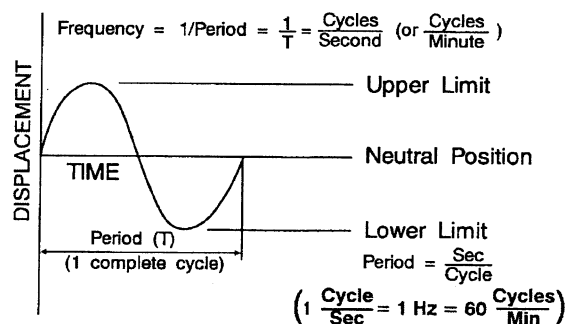
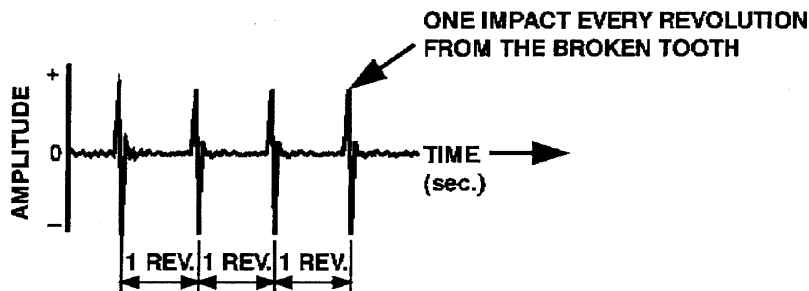


FIGURE 2
DISPLACEMENT AND FREQUENCY FROM A TIME WAVEFORM

When is a good time to use time waveforms in an analysis? Time waveforms are an excellent analytical tool to use when analyzing gearboxes. The transducer can be attached close to the input or the output shaft bearing to check for broken or chipped gear teeth. The following is a typical example of how a display for one broken tooth would appear as a time waveform, shown in Figure 3.



HERE IS HOW IT WOULD TYPICALLY BE DISPLAYED IN THE FREQUENCY DOMAIN:

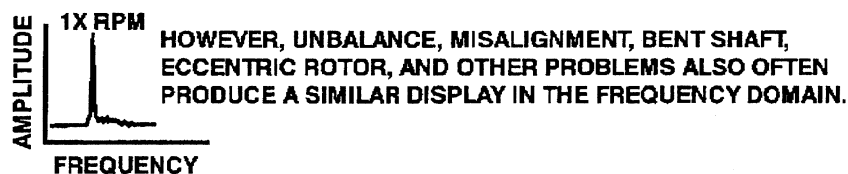


FIGURE 3
HOW A BROKEN TOOTH ON A GEAR IS DISPLAYED IN TIME WAVEFORM AND IN A SPECTRUM

Time waveforms are especially ideal for low-speed shafts and gears, even if some never rotate a full revolution (basically just rocking back and forth). In this case, time waveforms are virtually the only analytical tool which can be effectively used.

In the time waveform shown in the above example, an analyst can calculate the frequency of the impact or the speed of the shaft even though the display is in the time domain. If the time between each impact was given as 5 milliseconds (.005 second), the frequency would be calculated as:

$$\begin{aligned}
 F &= \frac{1}{T} \\
 &= \frac{1}{.005 \text{ SECOND/CYCLE}} \\
 &= 200 \text{ CYCLES/SECOND} \\
 &= 200 \text{ Hz} \\
 &= 200 \text{ CYC/SEC} \times 60 \text{ SEC/MIN} \\
 &= 12,000 \text{ CPM}
 \end{aligned}$$

Thus, the frequency of the impacts (or the speed of the shaft in this case) is 12,000 CPM. Likewise, it can be readily seen that if the time between impacts was 5 seconds instead, the frequency would only be .20 Hz ($1/5 = .20 \text{ cyc/sec}$) or 12 CPM - a very low frequency indeed. All this can be determined from a time waveform.

2.13 WHAT IS VIBRATION AMPLITUDE?

2.131 What is Vibration Displacement?

Displacement is a measure of the total travel of the mass - back and forth. Displacement can be expressed in mils (where 1 mil = .001 inch, or in microns (where 1 micron, μ = .001 millimeter or .039 mil). When a machine is being subjected to excessive dynamic stress at very low frequencies, displacement may be a good indicator of vibration severity since the machine (or structure) may be flexing too much; or simply being bent too far.

2.132 What is Vibration Velocity?

The velocity of the vibration is a measure of the speed at which the mass is moving or vibrating during its oscillations. The faster a machine flexes, the sooner it will fail in fatigue. **Vibration velocity is directly related to fatigue.** Note from the example of the oscillating mass suspended from a spring in Figure 4, that velocity reaches its maximum value (or peak) at the neutral position where the mass is fully accelerated (acceleration is zero) and now begins to decelerate as shown in Figure 4. Velocity is expressed as inches per second (in/sec) or as millimeters per second (mm/sec).

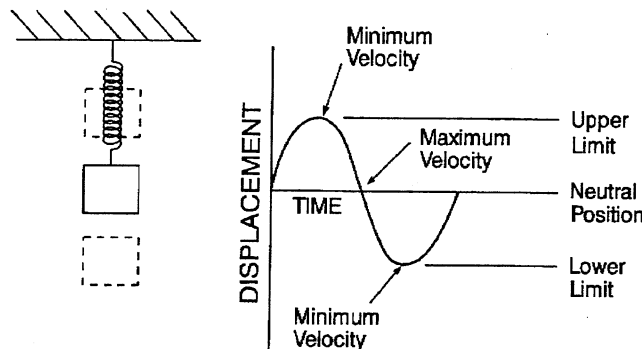


FIGURE 4
VELOCITY FROM THE DISPLACEMENT CURVE

However, if an analyzer was used to directly measure peak velocity, it would select the highest peak or excursion that the velocity time waveform would make. From an oscilloscope display, the peak velocity would be the highest peak in the display as shown in Figure 5.

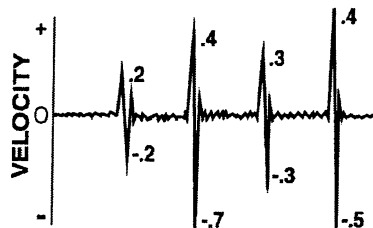


FIGURE 5
HOW TO DETERMINE PEAK VELOCITY FROM AN
OSCILLOSCOPE DISPLAY

In this case, the peak velocity is .7 in/sec because it is the highest peak, positive or negative.

2.133 What is Vibration Acceleration?

When a machine housing vibrates, it experiences acceleration since it continually changes speed as it oscillates back and forth. Acceleration is greatest at the instant at which velocity is at its minimum. That is, this is the point where the mass has decelerated to a stop and is about to begin accelerating (moving faster) in the opposite direction. Acceleration is the rate of change in velocity and is measured in units of g's (where $1\text{ g} = 32.2\text{ ft/sec}^2 = 386\text{ in/sec}^2 = 22.0\text{ mi/hr per second change}$). The greater the rate of change of velocity, the higher will be the forces (and stresses) on this machine due to the higher rate of acceleration. At high frequencies, failure of a machine may result from excessive forces which break down the lubrication allowing surface failures of bearings (due to metal-to-metal contact). These excessive forces are directly proportional to acceleration ($F=ma$). Acceleration is probably the most difficult measure of vibration amplitude to grasp, but is the parameter most often directly measured in the field with the use of an accelerometer. Thus, it is important that an analyst gain a good understanding of it.

2.14 WHAT IS VIBRATION PHASE?

Phase is a measure of how one part is moving (vibrating) in relation to another part, or to a fixed reference point. Vibration phase is measured in angular degrees by using either a strobe light or an electronic photocell. Figure 6 shows two masses vibrating with a 90° phase difference. That is, Mass #2 is one-fourth of a cycle (or 90°) ahead of Mass #1; thus, Mass #2 is "leading" Mass #1 in phase by 90° . Or, from the other point of view, Mass #1 has a 90° phase lag relative to the motion of Mass #2.

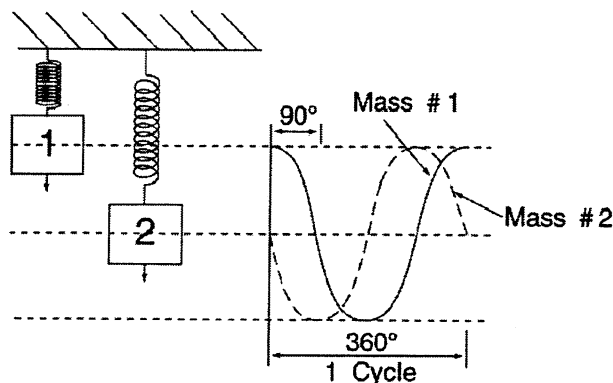


FIGURE 6
TWO MASSES WITH 90° PHASE DIFFERENCE

Figure 7 shows the same two masses vibrating with an 180° phase difference. That is, at any instant in time, Mass #1 will move downwards at the same instant as Mass #2 moves upwards, and vice versa.

Figure 8 shows how phase relates to machine vibration. The left sketch shows a 0° phase difference between bearing Positions 1 and 2 (in-phase motion). The right sketch shows a 180° out-of-phase difference between these positions (out-of-phase motion).

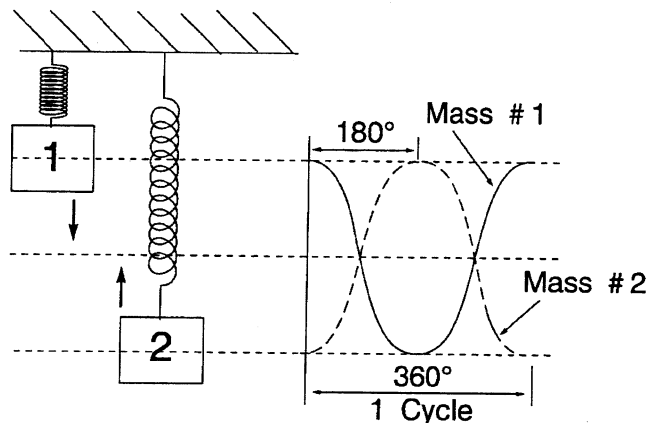


FIGURE 7
TWO MASSES WITH 180° PHASE DIFFERENCE

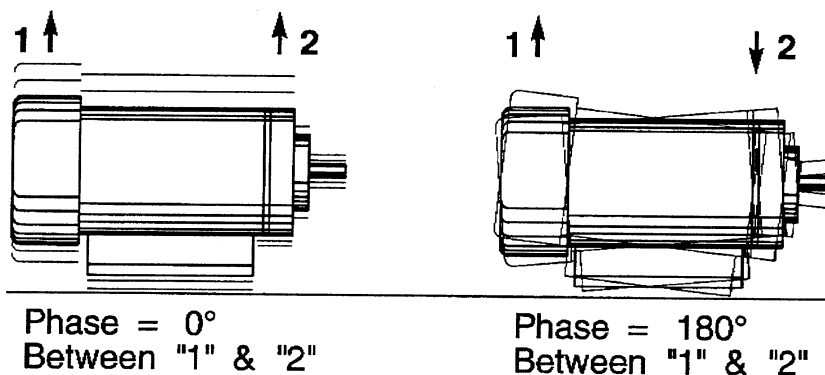


FIGURE 8
PHASE RELATIONSHIP AS USED WITH MACHINERY VIBRATION

2.141 How To Read Phase on CRT or RTA Screens

- a. At the dashed lines, the following illustrations of various time waveforms show that the same position on each wave maintains the same phase relationship. That is, for 90° or any other angle on each wave, the 90° location (or any other angle) remains the same on all waves regardless of how the waveform is displayed (that is, the location of 90° is at the highest positive point; 180° is at zero amplitude with the waveform sloping downwards; 270° is at the lowest negative point; while 0° (or 360°) is back at zero amplitude, but with the waveform sloping upwards (positive slope).

Another point to be made about the waveforms of Figure 9 is to show how waveforms can be used to compare phase at various locations. For example, Waveform A might be at the driver outboard bearing horizontal; Waveform B at the driver inboard bearing horizontal; while Waveform C is at the inboard bearing of the driven machine. If all 3 waveforms were captured simultaneously, phase comparisons can be made. In this case, it would show there is a 180° phase difference between Waveforms A and B (when A goes up, B goes down and vice versa). On the other hand, there is only a 90° phase difference between Waveforms A and C.

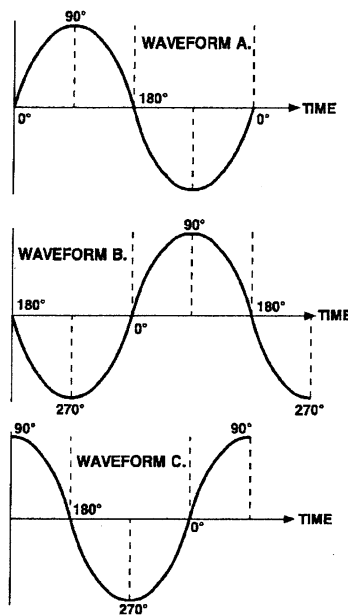
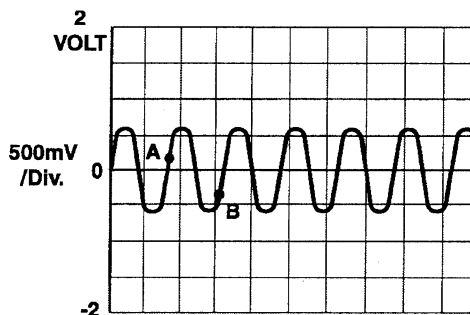


FIGURE 9
HOW TO DETERMINE THE PHASE DIFFERENCE
BETWEEN TWO TIME WAVEFORMS

b. How to determine the phase difference between two points on the same time waveform:



PROBLEM: WHAT IS THE PHASE DIFFERENCE (SHAFT)
 BETWEEN POINT A AND POINT B ON THE
 DISPLAY, IN DEGREES ?

SOLUTION 1

- 1) PHASE FOR POINT A = 30°
- 2) PHASE FOR POINT B = 320°
- 3) THE DIFFERENCE IS:
 FROM $30^\circ \rightarrow 90^\circ = 60^\circ$
 FROM $90^\circ \rightarrow 270^\circ = 180^\circ$
 FROM $270^\circ \rightarrow 320^\circ = 50^\circ$
 ADD 290°

SOLUTION 2

- 1) COUNT THE NUMBER OF PEAKS ACROSS THE 10 DIVISIONS.
 THERE ARE APPROXIMATELY 6.4 PEAKS.
 THERE ARE APPROXIMATELY 6.4 CYCLES.
- 2) DIVIDE 6.4 CYCLES INTO 10 DIVISIONS.
 THERE ARE $\frac{10}{6.4} = 1.56$ DIVISIONS
 CYCLES
- 3) MEASURE THE DISTANCE IN DIVISIONS BETWEEN POINT A POINT B.
 THERE ARE APPROXIMATELY 1.25 DIVISIONS.
- 4) THERE ARE 360° IN ONE CYCLE,
 THERE ARE $\frac{1.25}{1.56} \times 360^\circ = 288^\circ$ (290°)
 DIFFERENCE BETWEEN POINTS A & B.

FIGURE 10
HOW TO DETERMINE THE PHASE DIFFERENCE BETWEEN TWO POINTS
ON THE SAME SINUSOIDAL TIME WAVEFORM

2.142 Phase Relationship of Acceleration, Velocity & Displacement Time Waveforms

Figure 11 shows the phase relationship between acceleration, velocity and displacement time waveforms. *It shows that acceleration leads velocity by 90° and leads displacement by 180°.* On the other hand velocity *lags* acceleration by 90°, but *leads* displacement by 90°. Finally displacement *lags* acceleration by 180° and *lags* velocity by 90°.

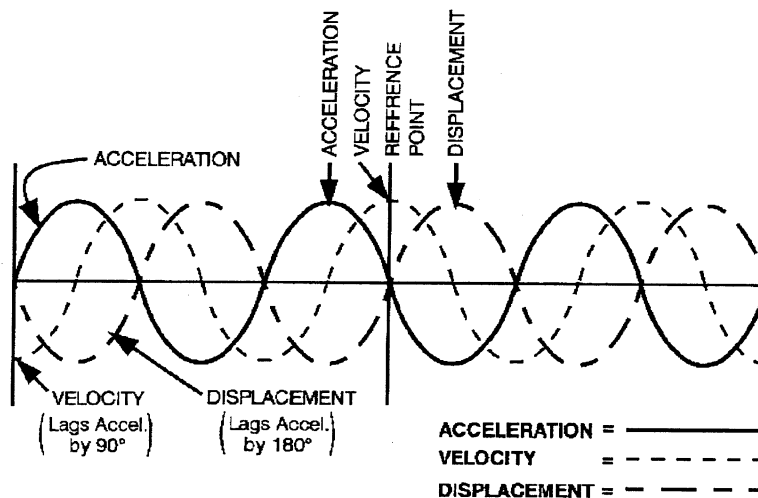


FIGURE 11
PHASE RELATIONSHIP BETWEEN ACCELERATION, VELOCITY AND DISPLACEMENT TIME WAVEFORMS

2.15 WHAT IS A VIBRATION SPECTRUM (ALSO CALLED AN "FFT" OR "SIGNATURE")?

Most vibrations in the real world are complex combinations of various waveforms. *Figure 12 shows how the total waveform is actually made up of a series of smaller waveforms*, each of which correspond to an individual frequency (1X RPM, 2X RPM, 3X RPM, etc.). *Each of these individual waveforms will algebraically add to one another to generate the total waveform* which can be displayed either on an oscilloscope or on an analyzer.

One of the most important points to understand about the total time waveform is that it shows the total vibration motion of the machine or structure to which the vibration transducer is attached. If one can begin to comprehend this point, examination of the time waveform can go far in helping him diagnose both the cause and severity of problem(s) occurring within a machine.

However, particularly when an analyst is just beginning within the field of vibration analysis (typically less than 3 years full-time experience), displaying and using the time waveform can be very difficult and labor intensive if one needs to determine frequencies. To simplify the process, a Fast Fourier Transform (FFT) is generated and displayed within most of today's vibration data collectors and spectrum analyzers. *An FFT is a computer (microprocessor) transformation from time domain data (amplitude versus time) into frequency domain data (amplitude versus frequency).*

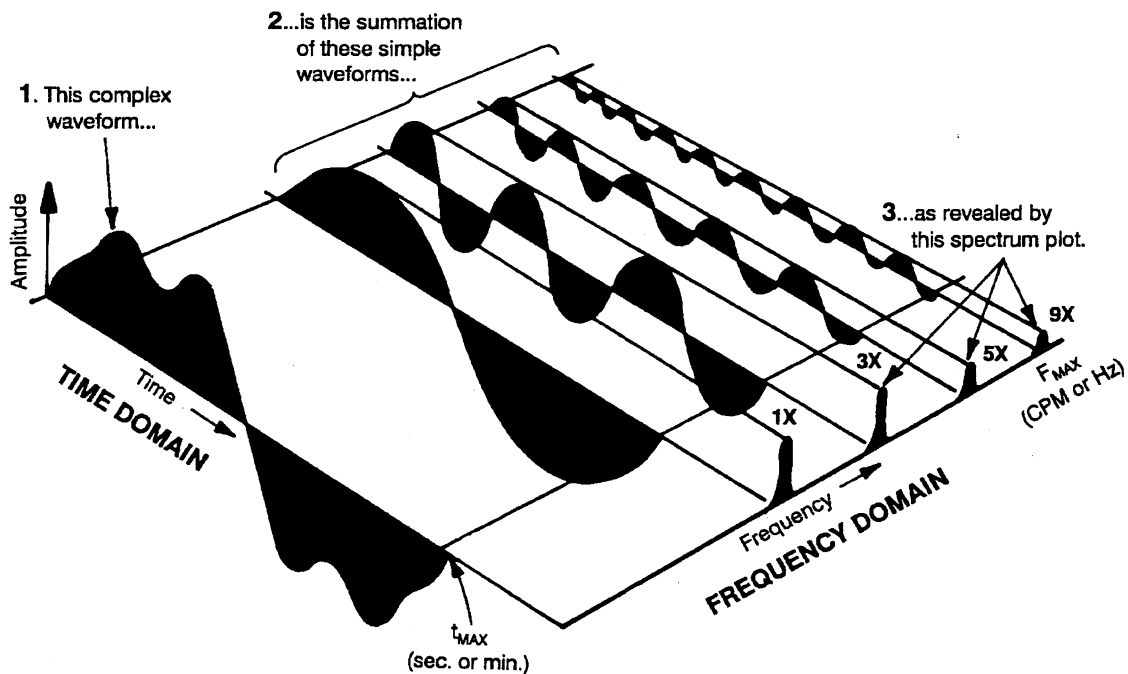


FIGURE 12
COMPARISON OF TIME & FREQUENCY DOMAINS

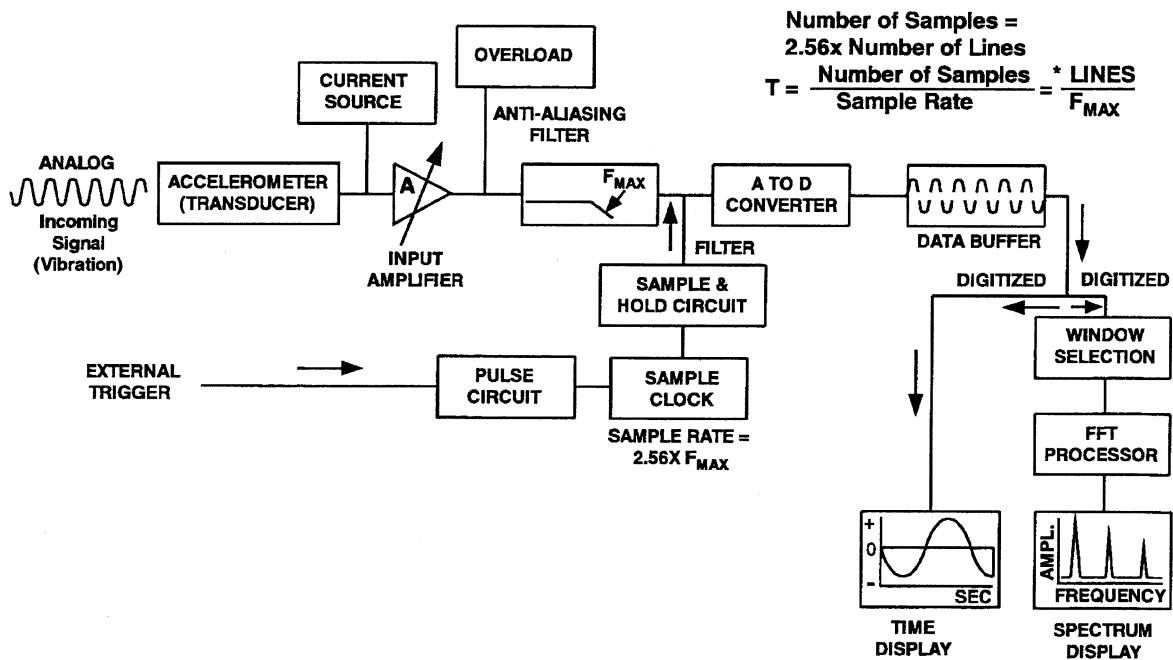


FIGURE 13
BLOCK DIAGRAM OF A GENERAL FFT ANALYZER TO SHOW HOW A DISPLAY IS PRODUCED IN EITHER THE TIME DOMAIN OR FREQUENCY DOMAIN

Figure 13 is a block diagram of a general FFT analyzer. Its purpose is to show how a digital time waveform or FFT spectrum is generated from the incoming analog vibration raw signal. The process on how an FFT spectrum is produced will be covered in later chapters. Remember that an FFT is a microprocessor algorithm (mathematical operation also used in computers) applied to the incoming sampled data from the signal captured in the analog world (time domain). This must be transformed into the frequency domain using a series of mathematical operations.

This FFT calculation technique was developed by Baron Jean Baptiste Fourier over 170 years ago (1822). Fourier proved that any real-world complex waveform can be separated into simple sinusoidal waveform components. The converse is also true: any series of simple sine waves can be combined to create the complex total waveform. As the sine waves are separated from the combined waveform, they are converted to vertical peaks which have an amplitude (as determined by their heights) and are given a position along the frequency axis. This frequency domain presentation of a time waveform is called a spectrum (spectra, plural). A spectrum is also sometimes referred to as a "signature" or as an "FFT" if an FFT analyzer is used.

Figure 14 summarizes the steps involved in capturing the total vibration waveform and transforming it into the frequency domain (FFT) as the signal is sent from a transducer mounted on the bearing housing in the real world. Of course, this figure shows the transformation of only one frequency. In the real world, machines will generate many frequencies from many sources within the machine. Diagnosing these spectra will be the main topics of the chapters in which the items in the information will be summarized in an "Illustrated Vibration Diagnostics Chart" developed by Technical Associates will be summarized.

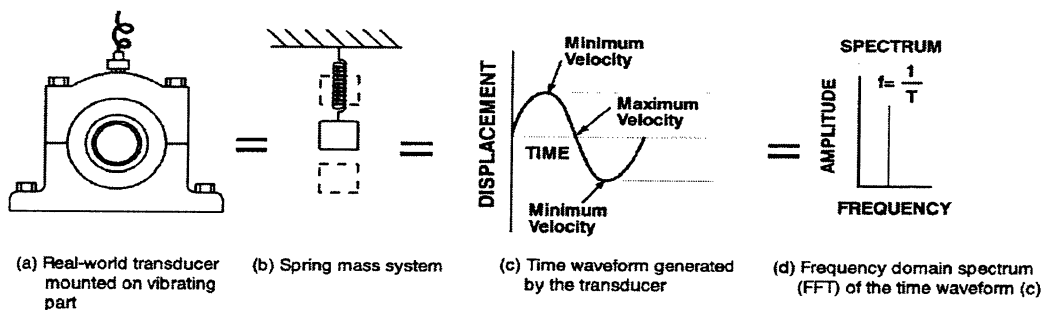


FIGURE 14
STEPS IN THE CONVERSION OF A VIBRATION INTO AN FFT SPECTRUM

2.16 DIFFERENCE BETWEEN RMS, PEAK AND PEAK-TO-PEAK AMPLITUDE

Table I is a list of formulas which can be used to convert from one amplitude parameter to another. That is, it allows one to convert from displacement to velocity at a certain frequency; or from velocity to acceleration, etc. Thus, if one parameter is a peak value which has been measured, then the parameter which is being calculated will also be a peak value.

TABLE I
CONVERSION FORMULAS FOR VARIOUS AMPLITUDE UNITS (Ref. 5)

ENGLISH	METRIC
$D = 19,100 (V)/F$	$D_M = 19,100 (V_M)/F$
$D = 70,471,000 (A)/F^2$	$D_M = 1.79 \times 10^9 (A)/F^2$
$V = DF/19,100$	$V_M = D_M F/19,100$
$V = 3690 (A)/F$	$V_M = 93,580 (A)/F$
$A = DF^2/70,471,000$	$A = D_M F^2/1.79 \times 10^9$
$A = VF/3690$	$A = V_M F/93,580$

A= Acceleration (g PK)

V= Velocity (in/sec PK)

D= Displacement (mils PK-PK)

F= Frequency (CPM)

A= Acceleration (g PK)

V= Velocity (mm/sec PK)

D= Displacement (microns PK-PK)

F= Frequency (CPM)

However, conversions from one vibration parameter to another are normally done by the software and electronics within the vibration instrument. Also, the electronics can perform all the necessary conversions for peak-to-peak, peak, and RMS (root-mean-square) amplitude values. Normally, Europeans use RMS velocity amplitudes, while Americans have adopted peak values even though, in reality, the instruments truly display RMS spectra, and then electronically convert them to so-called "peak" or "peak-to-peak" spectra by multiplying each of the amplitudes of each of the frequencies by $1.414 (\sqrt{2})$ in the case of RMS-to-Peak; or by 2.828X in the case of RMS to "peak-to-peak" (assuming such waveform is sinusoidal). Figure 15 compares the English vibration units with the Metric.

VIBRATION CHARACTERISTICS	COMMON UNITS OF MEASUREMENT	
	ENGLISH	METRIC
FREQUENCY	CPM OR Hz	CPM OR Hz
DISPLACEMENT	MILS PEAK-TO-PEAK	MIICRONS, PEAK
VELOCITY	IN/SEC RMS IN/SEC PEAK	MM/SEC RMS MM/SEC PEAK
ACCELERATION	G PEAK	G RMS
PHASE	DEGREES	DEGREES

FIGURE 15
COMPARISON OF ENGLISH AND METRIC VIBRATION UNITS (Ref. 5)

Figure 16 shows how one unit of amplitude can be converted to another; that is, from RMS to peak, peak-to-peak, and vice versa. These conversions apply only to pure sinusoidal waves only (likely caused by almost pure unbalance), similar to the waveform shown in Figure 10.

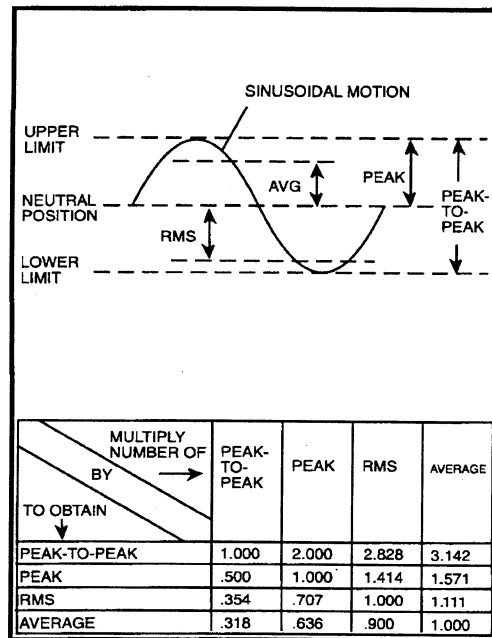


FIGURE 16
COMPARISON OF PEAK, PEAK-TO-PEAK, RMS, AND AVERAGE FOR A
PURE SINUSOIDAL TIME WAVEFORM

2.17 WHEN TO USE DISPLACEMENT, VELOCITY, OR ACCELERATION

Displacement is normally thought to be the most useful vibration parameter in frequency ranges less than approximately 600 CPM (10 Hz). However, a frequency must be used along with displacement to evaluate vibration severity as shown by Figure 17. For instance, 2 mils Pk-Pk of vibration at 3600 CPM is much more destructive than is the same 2 mils vibration at 300 CPM (see Figure 17 which is a displacement and velocity severity chart developed years ago for "general rotating machines"). Thus, displacement alone is unable to evaluate vibration severity throughout the entire frequency range (even for a low-speed machine).

Acceleration is also frequency dependent (see Figure 18 which is a vibration severity chart for acceleration). Typically, acceleration is recommended for use when sources within a machine generate frequencies over approximately 120,000 CPM (2000 Hz). These sources may include gear mesh frequencies (#teeth X RPM) and blade passing frequencies (#blades X RPM) for high-speed centrifugal as well as harmonics (or multiples) of these frequencies.

On the other hand, velocity is not nearly so frequency dependent in the frequency range from approximately 600 - 120,000 CPM (10 - 2000 Hz). Even when vibration frequencies are generated from 300 to as high as 300,000 CPM in a machine, velocity is usually the unit of choice (although one will have to take into account the roll-off in sensitivity of velocity at frequencies exceeding 120,000 CPM as shown in Figure 19). For example, if one allowed a velocity of .10 in/sec at 120,000 CPM for a fault such as a gear mesh frequency, he would likely allow only a level of about .04 in/sec at a frequency of 300,000 CPM $[(120,000/300,000)(.10) = .04]$ as per the equations and graphs identified as "CONTOURS OF EQUAL SEVERITY" shown in Figure 19. For the same reason, if one allowed a velocity of .314 in/sec. at 600 CPM, he should allow a level of only .031 in/sec. at a frequency 60 CPM due to fall off in velocity below 600 CPM (10Hz) as shown by Figure 19. Figure 19 shows that if he still allowed .314 in/sec. at 60 CPM, this would be

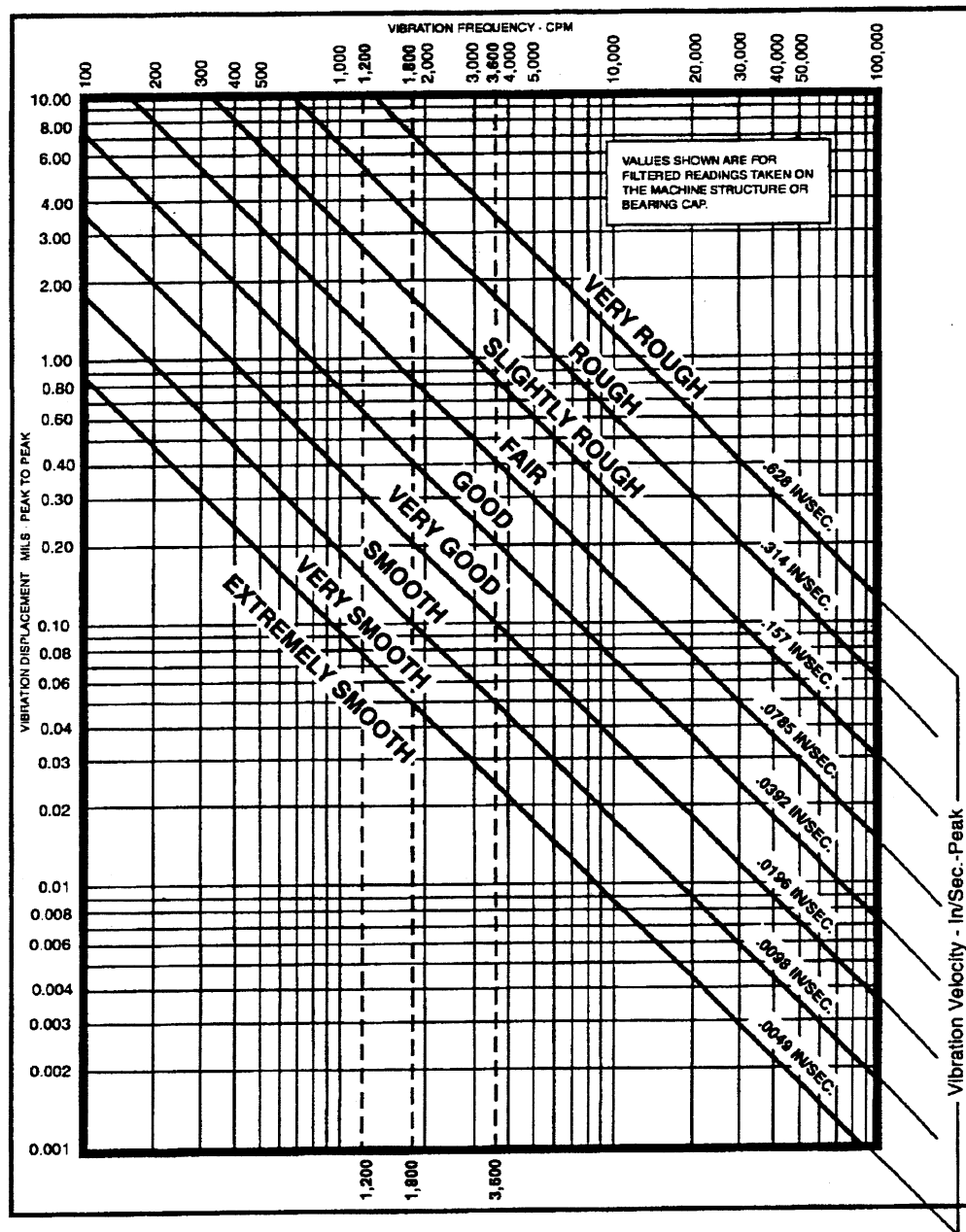


FIGURE 17
VIBRATION DISPLACEMENT & VELOCITY SEVERITY CHART FOR
GENERAL HORIZONTAL ROTATING MACHINERY
 (Source: Entek IRD International, Milford, Ohio)

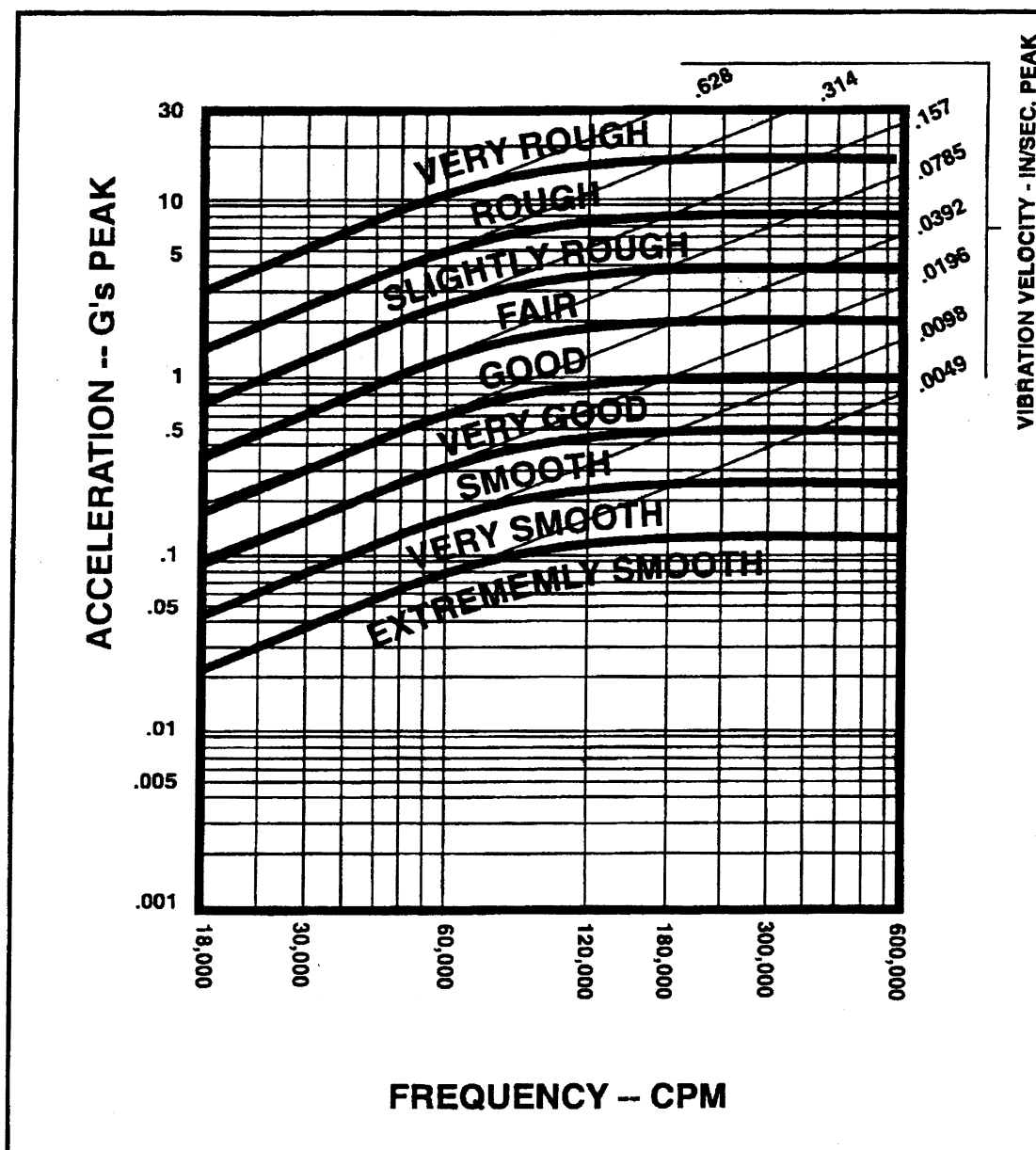


FIGURE 18
VIBRATION ACCELERATION & VELOCITY SEVERITY CHART FOR
GENERAL HORIZONTAL ROTATING MACHINERY
(Source: Entek IRD International, Milford, Ohio)

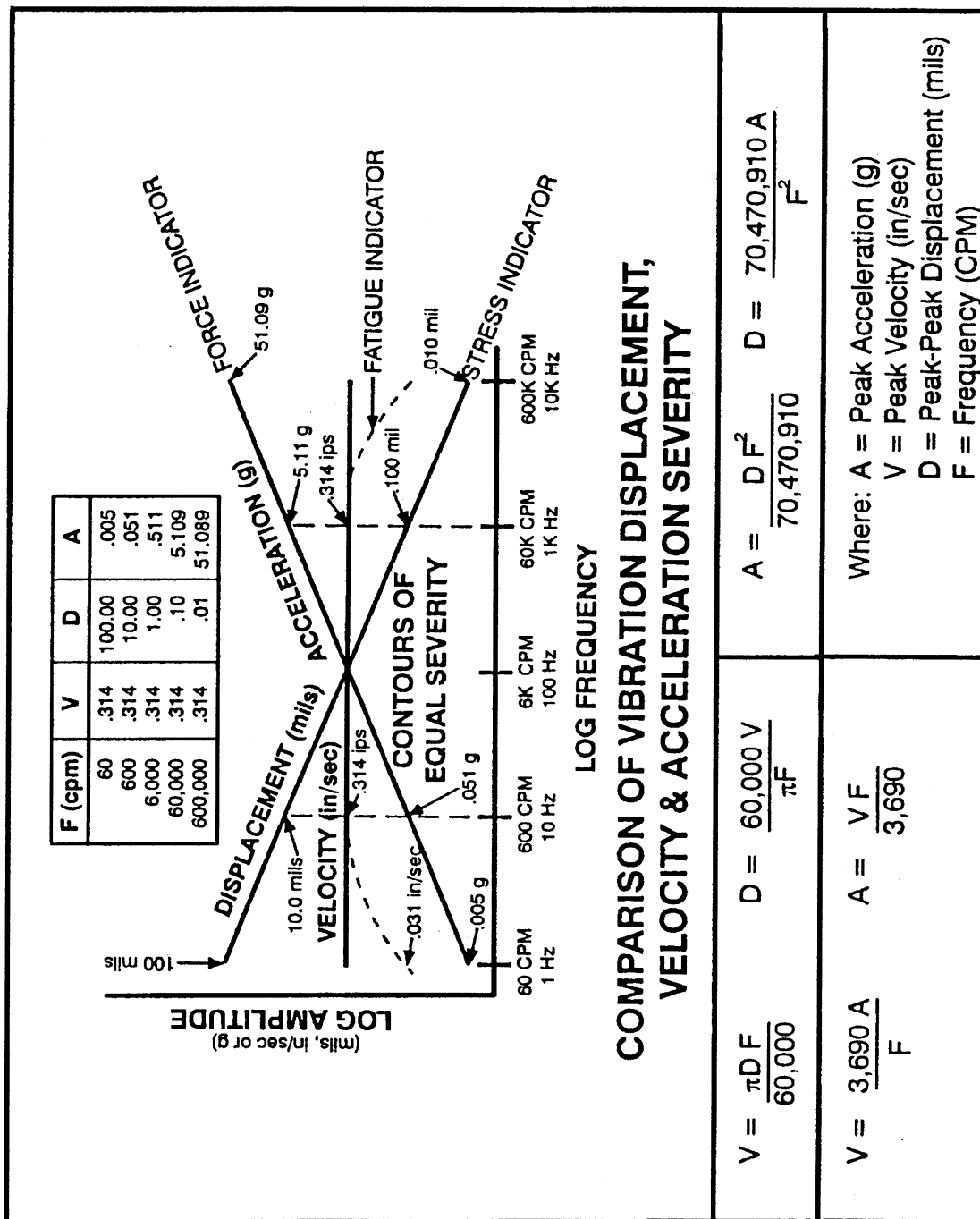
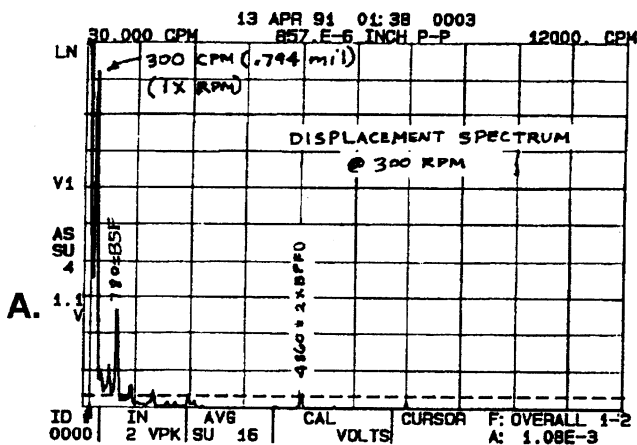
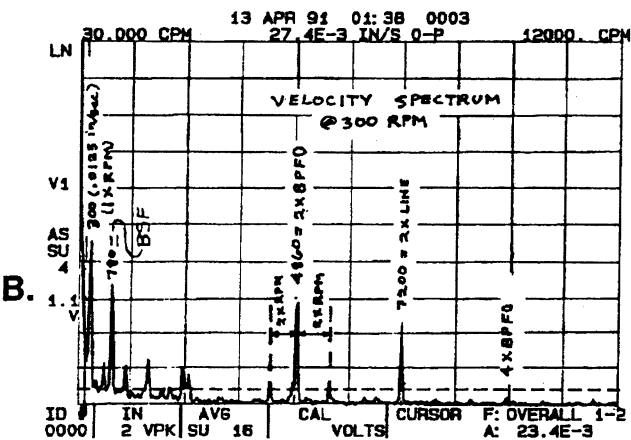


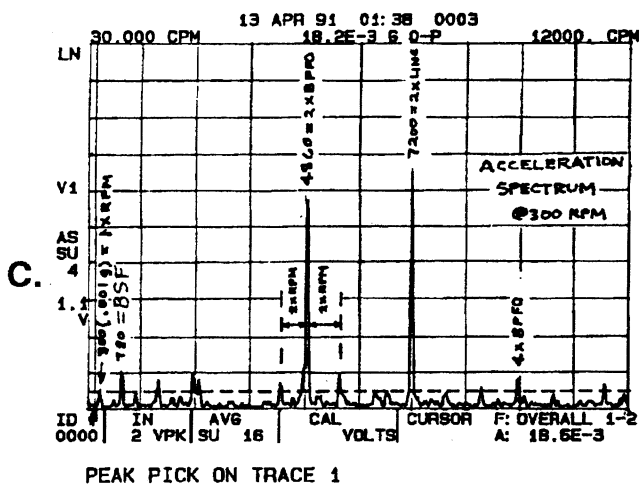
FIGURE 19



INCH P-P	CPM
794.E-6	300.00
82.2E-6	420.00
97.8E-6	600.00
227.E-6	780.00
50.9E-6	1110.0
39.1E-6	1589.9
31.3E-6	4860.0



IN/S O-P	CPM
12.5E-3	300.00
1.75E-3	420.00
3.13E-3	600.00
9.26E-3	780.00
1.13E-3	900.00
2.88E-3	1110.0
1.13E-3	1500.0
3.38E-3	1589.9
1.13E-3	1890.0
1.25E-3	2070.0



G O-P	CPM
997.E-6	300.00
1.91E-3	780.00
1.50E-3	1589.9
1.75E-3	2340.0
1.50E-3	2490.0
1.33E-3	4260.0
10.5E-3	4860.0
1.83E-3	5579.7
914.E-6	6360.0
914.E-6	6629.7
11.9E-3	7200.0

FIGURE 20
COMPARISON OF DISPLACEMENT, VELOCITY & ACCELERATION
SPECTRA ON A 300 RPM FAN WITH BEARING PROBLEMS

equivalent to an excessive displacement of 100 mils (PK-PK). Velocity can still be used at low speeds like 60 CPM, since bearing defect frequencies, gear mesh frequencies and their harmonics will still normally be higher than 600 CPM (10Hz), which is considered the "break point" between low speed analysis and moderate speed analysis techniques.

2.171 What is the Advantage of Using Velocity?

Figure 19 shows the consistency which velocity has over a wide, flat frequency range as compared with displacement and acceleration. They tend to favor the low and the high ends of the frequency scale, respectively. Note in Figure 19 that all 3 amplitude parameters are displayed on the same graph, using the .314 in/sec peak velocity amplitude as a basis for the calculation of the "CONTOURS OF EQUAL SEVERITY".

The following example, as displayed in Figure 20, shows 3 spectra in (A) displacement, (B) velocity, and (C) acceleration of the same waveform. Carefully analyze these spectra for a possible bearing defect problem. Although the 1X RPM peak (300 CPM) appears in all three spectra and is even quite outstanding in Figure 20A, it is not the most significant problem here.

What happens to the display of bearing defect frequencies if using one vibration parameter or another is possibly more important when one is evaluating machine condition. Whether or not the analyst will see these important bearing frequencies in his spectra may depend upon his choice of amplitude parameter. While bearing frequencies at 4860 CPM and 9720 CPM with their sideband frequencies are clearly seen in Figures 20B and 20C, note that the frequency at 9720 CPM is missed entirely in the displacement spectrum, as well as the sidebands surrounding the 4860 CPM frequency in Figure 20A. This is very important. If an acceleration spectrum in Figure 20C wasn't taken, these frequencies still showed up significantly in the Figure 20B velocity spectrum. If the displacement spectrum in Figure 20A wasn't taken, the 1X RPM spike was still significant in the velocity spectrum of Figure 20B. Therefore, if only a velocity spectrum had been taken, as in Figure 20B, both types of problems would be clearly visible.

Thus, it is important to note that a velocity spectrum has a much wider usable frequency range than do spectra in displacement or acceleration. Combining this characteristic with velocity's direct relationship to vibration severity makes velocity the best measurement parameter to use for most rotating machinery. This is especially true when frequencies are below 120,000 CPM (2000 Hz).

2.18 HOW MUCH IS TOO MUCH VIBRATION?

Through the years, the general vibration severity chart of Figure 17 has been commonly used. However, this chart was never intended to be used on all machine types and configurations to choose vibration limits to give adequate warning of existing or impending problems. To help meet this need, Technical Associates has developed a comprehensive vibration severity chart, shown in Figure 21, which is entitled "CRITERIA FOR OVERALL CONDITION RATING". This chart applies to a wide variety of machines over a wide range of operating speeds from 600 - 60,000 RPM. These levels are peak overall velocity levels (in/sec). They were acquired through many years of actual vibration data acquisition on a diverse array of machine types. The columns entitled "GOOD" and "FAIR" are used to give a machine an "OVERALL CONDITION RATING" based on the highest overall vibration level found on any one of the machine measurement points. In general, machines allowed to operate above "ALARM 1" will likely fail prematurely if problems are not identified and corrected. "ALARM 2" levels are 50% higher than those of "ALARM 1". If machines are allowed to operate above "ALARM 2", they may suffer catastrophic failure if left unaddressed.

This Technical Associates Rating Chart does not cover all types of machines. Further, it is meant for "IN-SERVICE" equipment only. It is not meant to be used for "ACCEPTANCE TESTING". For machines not included in the chart of Figure 21, one could use the Figure 17 severity chart or a statistical method to develop other alarm levels. A statistical comparison can be conducted if the machines are similar in construction, drive configuration, operating speeds, loading and in internal components. This statistical method is especially effective when several surveys on the machines have been conducted. It is practical to revise the alarms since the original vibration levels are almost always reduced as machine problems and defects are corrected.

Shock Pulse, HFD, and Spike Energy, as other measurement parameters, will be discussed later in several other chapters.

FIGURE 21

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

CRITERIA FOR OVERALL CONDITION RATING (PEAK OVERALL VELOCITY, IN/SEC)*

1. Assuming Machine Speed = 600 to 60,000 RPM.
2. Assuming Measurements by Accelerometer or Velocity Pickup securely mounted as Close as Possible to Bearing Housing.
3. Assuming Machine Is Not Mounted on Vibration Isolators (for Isolated Machinery - Set Alarm 30% to 50% Higher).
4. Set Motor Alarms the Same as that for the Particular Machine Type unless Otherwise Noted.
5. Consider Setting Alarms on Individual External Gearbox Position about 25% Higher than that for a particular Machine Type.

MACHINE TYPE	GOOD	FAIR	(WARNING)	(FAULT)
			ALARM 1	ALARM 2
COOLING TOWER DRIVES				
Long, Hollow Drive Shaft	0 - .375	.375 - .600	.600	.900
Close Coupled Belt Drive	0 - .275	.275 - .425	.425	.650
Close Coupled Direct Drive	0 - .200	.200 - .300	.300	.450
COMPRESSORS				
Reciprocating	0 - .325	.325 - .500	.500	.750
Rotary Screw	0 - .300	.300 - .450	.450	.650
Centrifugal With or W/O External Gearbox	0 - .200	.200 - .300	.300	.450
Centrifugal - Integral Gear (Axial Meas.)	0 - .200	.200 - .300	.300	.450
Centrifugal - Integral Gear (Radial Meas.)	0 - .150	.150 - .250	.250	.375
BLOWERS (FANS)				
Lobe-Type Rotary	0 - .300	.300 - .450	.450	.675
Belt-Driven Blowers	0 - .275	.275 - .425	.425	.650
General Direct Drive Fans (with Coupling)	0 - .250	.250 - .375	.375	.550
Primary Air Fans	0 - .250	.250 - .375	.375	.550
Vacuum Blowers	0 - .200	.200 - .300	.300	.450
Large Forced Draft Fans	0 - .200	.200 - .300	.300	.450
Large Induced Draft Fans	0 - .175	.175 - .275	.275	.400
Shaft-Mounted Integral Fan (Extended Motor Shaft)	0 - .175	.175 - .275	.275	.400
Vane-Axial Fans	0 - .150	.150 - .250	.250	.375
MOTOR/GENERATOR SETS				
Belt-Driven	0 - .275	.275 - .425	.425	.675
Direct Coupled	0 - .200	.200 - .300	.300	.450
CHILLERS				
Reciprocating	0 - .250	.250 - .400	.400	.600
Centrifugal (Open-Air) - Motor & Compressor Separate	0 - .200	.200 - .300	.300	.450
Centrifugal (Hermetic) - Motor & Impellers Inside	0 - .150	.150 - .225	.225	.350
LARGE TURBINE/GENERATORS				
3600 RPM Turbine/Generators	0 - .175	.175 - .275	.275	.400
1800 RPM Turbine/Generators	0 - .150	.150 - .225	.225	.350
CENTRIFUGAL PUMPS				
Vertical Pumps (12' - 20' Height)	0 - .325	.325 - .500	.500	.750
Vertical Pumps (8' - 12' Height)	0 - .275	.275 - .425	.425	.650
Vertical Pumps (5' - 8' Height)	0 - .225	.225 - .350	.350	.525
Vertical Pumps (0' - 5' Height)	0 - .200	.200 - .300	.300	.450
General Purpose Horizontal Pump - Direct Coupled	0 - .200	.200 - .300	.300	.450
Boiler Feed Pumps - Horizontal Orientation	0 - .200	.200 - .300	.300	.450
Piston Type Hydraulic Pumps - Horizontal Orientation (under load)	0 - .150	.150 - .250	.250	.375
MACHINE TOOLS				
Motor	0 - .100	.100 - .175	.175	.250
Gearbox Input	0 - .150	.150 - .225	.225	.350
Gearbox Output	0 - .090	.090 - .150	.150	.225
Spindles:				
a. Roughing Operations	0 - .065	.065 - .100	.100	.150
b. Machine Finishing	0 - .040	.040 - .060	.060	.090
c. Critical Finishing	0 - .025	.025 - .040	.040	.060

*NOTE: The "ALARM 1" and "ALARM 2" overall levels given above apply only to In-service machinery which has been operating for some time after initial installation and/or overhaul. They do not apply (and are not meant to serve as) Acceptance Criteria for either new or rebuilt machinery.

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2.2 OVERVIEW OF THE STRENGTHS AND WEAKNESSES OF TYPICAL VIBRATION INSTRUMENTS

2.21 INTRODUCTION

The purpose of vibration instrumentation is to accurately measure vibration amplitudes, frequencies, and phase so that a reliable determination of a machine's condition can be made. There are 5 basic types of vibration instruments as follows:

- 1) **Overall Level Vibration Meters**
- 2) **Swept-Filter Analyzers**
- 3) **FFT Programmable Data Collectors**
- 4) **Real-Time Spectrum Analyzers**
- 5) **Instrument Quality Tape Recorders**

2.22 INSTRUMENT COMPARISONS

This section is meant to evaluate the general capabilities of the 5 types of instruments listed above. It is important to note that not all makes and models of these instrument types which are configured to exhibit each of these characteristics that will be featured. However, this section provides a good checklist to review with an instrument manufacturer to fully understand the instrument's capabilities.

Table II will present each type of instrument and list the general capabilities each possesses. The comparison characteristics will be defined in more detail below only if they are somewhat complex.

- A) **Portability** - Can the equipment be easily carried around the plant or mill? How much does it weigh?
- B) **Typical Frequency Range** - Describes the typical range of frequencies from a low limit to a high limit in which an instrument of each particular type can accurately measure according to a specified amplitude tolerance (usually $\pm 10\%$ or $\pm 3\text{dB}$).
- C) **Data Measurement Format** -
 - 1) OL (Overall Level)
 - 2) SF (Swept Filter)
 - 3) FS (Frequency Spectrum)
 - 4) TWF (Time Waveform)
- D) **Typical Display Types** -
 - 1) LCD (Liquid Crystal Display)
 - 2) MS (Monochrome Screen)
 - 3) AM (Analog Meter)
- E) **Typical Transducer Types** -
 - 1) A (Accelerometer)
 - 2) V (Velocity Transducer)
 - 3) P (Proximity Eddy Current Probe)

-
- F) **Phototach and/or Strobe Light Capabilities** - Can the instrument type normally use a Phototach or strobe light with which it can measure phase, as well as enabling it to possibly perform operating deflection shape, modal analysis and/or synchronous time averaging?
- G) **Multi-Channel Availability** - Is this instrument type typically available in more than one channel?
- H) **Spike Energy, HFD, or SPM (Shock Pulse Measurement) Capability** - Can the instrument type typically measure one of these parameters?
- I) **High Frequency Enveloped Spectral Measurement Capability** - High frequency enveloped spectra are known by different vendors as "Spike Energy Spectra"¹¹, "Amplitude Demodulated Spectra"¹², or "Acceleration Enveloped Spectra"¹³, which are usually measured in the 5000 - 50,000 Hz (300,000 - 3,000,000 CPM) frequency range. However, the "SEE" spectrum¹⁴ developed by SKF Condition Monitoring is measured in the 250,000 - 350,000 Hz (15,000,000 - 21,000,000 CPM) range (where SEE refers to Spectral Emitted Envelope). These parameters will be covered in later chapters of the text.
- J) **Spectral Display Update** - How fast does the screen refresh itself with up-to-date data?
- **LT (Live Time)** - Screen updates every 1 to 4 seconds depending on the instrument model and the settings, such as the frequency span, the number of lines of resolution, the overlap processing percentage, etc.
 - **RT (Real Time)** - Screen updates almost instantaneously, particularly in higher frequency spans (again depends on instrument setup parameters just as on a data collector).
- K) **Ease of Use** - An assessment rating from simple to complex based on the time and training normally required to operate the instrument effectively. The assessment has to include a consideration of whether the instrument will be used regularly (daily, weekly) or occasionally (monthly).
- L) **Time Waveform Storage Capability** - Can this instrument type typically acquire and store time waveform?
- M) **Frequency Spectra Storage Capability** - Can this instrument type typically acquire and store frequency spectra?
- N) **Predictive Maintenance (PMP) Software Compatibility** - Is the instrument compatible with available condition monitoring software to set up overall and spectral alarms, trend data, routes, etc.?
- O) **Natural Frequency Testing Capability** - Can the instrument be used to conduct "bump" or "impulse" tests, coastdown/runup tests, Bode' or Polar plot measurements?
- P) **ODS (Operating Deflection Shape) Capability** - The ability to simultaneously measure the amplitude and phase at a particular forcing frequency (such as 1X or 2X RPM) are measured at specified locations on a structure or machine and typically are downloaded into a personal computer. Software in the computer is designed to produce animated operating deflection shape plots on the screen. This will simulate how the remainder of the machine or structure is moving in relation to one of the points. This can be accomplished using a single channel analyzer, along with a once/revolution trigger.

-
- Q) **Experimental Modal Analysis Capability** - The capability to measure items required by modal analysis such as natural frequencies, mode shapes, coherence and transfer functions. Modal analysis involves exciting the natural frequencies of the structure with the use of a "modal hammer" (force transducer) or by a shaker likewise outfitted with a force transducer, and measuring the response with an accelerometer. This analysis requires at least a two channel analyzer.
 - R) **STA (Synchronous Time Averaging) Capability** - The capability to eliminate all frequencies that are not exact multiples of a designated frequency. The spectrum being measured will be limited to only multiples of the fundamental frequency (most often, operating speed) which is synchronous with the trigger source (such as a phototach or strobe light). The nonsynchronous frequencies will disappear from the spectrum and time waveform if a sufficient number of averages are taken (often 250 to 500 averages).
 - S) **Waterfall or Cascade Plotting Capability** - The capability to display one FFT after another during a "runup" or "coastdown" and/or from one PMP survey to the next on the screen.
 - T) **Relative Costs** - From a basic instrument "low-end" cost, the range may vary considerably to the "high-end" cost, depending on the software, cabling, number of channels, the auxiliary equipment, and other "extras" to be purchased. The "nominal cost" represents what is normally paid for these instruments and the necessary "extras".

TABLE II
TYPICAL VIBRATION MEASUREMENT INSTRUMENT
CHARACTERISTICS @ 6/93

CHARACTERISTICS	OVERALL LEVEL METERS	SWEPT- FILTER ANALYZERS	FFT DATA COLLECTORS	REAL- TIME ANALYZERS	TAPE RECORDERS
A. PORTABILITY Approximate Weight (Lb)	YES 1-2	SEMI 25-45	YES 5-10	SEMI 25-60	SEMI 20-40
B. TYPICAL FREQUENCY RANGE (CPM) Low End High End	120 600,000	60 600,000	60 2,400,000	0 6,000,000	VARIES VARIES
C. DATA MEASUREMENT FORMAT	OL	OL, SF	OL, FS, TWF	OL, FS, TWF	-
D. TYPICAL DISPLAY TYPES	AM	AM	LCD	MS TYPICALLY	-
E. TYPICAL TRANSDUCER TYPES	PER MANUF	A, V	A, V, P	A, V, P	A, V, P
F. PHOTOTACH AND/OR STROBE LIGHT	NO	YES	MOST USE BOTH	PTACH-YES STROBE-NO	PTACH-YES STROBE-NO
G. MULTI-CHANNEL AVAILABILITY	NO	LIMITED	SOME	YES	YES
H. SPIKE ENERGY, HFD, OR SHOCK PULSE	SOME	SOME	MOST	NO	NO
I. HIGH FREQUENCY ENVELOPED SPECTRUM	NO	NO	MOST	FEW	NO
J. SPECTRAL DISPLAY UPDATE	-	-	LT	RT	-
K. EASE OF USE	SIMPLE	SIMPLE	FAIRLY SIMPLE	DEMANDING	VARIES
L. TIME WAVEFORM STORAGE CAPABILITY	NO	HC ONLY	MOST	YES	YES
M. FREQUENCY SPECTRA STORAGE	NO	NO	YES	YES	-
N. PREDICTIVE MAINT. SOFTWARE AVAILABILITY	NO	NO	YES	SOME	-
O. NATURAL FREQUENCY TESTING CAPABILITY	NO	LIMITED	GOOD	SUPERIOR	-
P. OPERATING DEFLECTION SHAPE (ODS)	NO	NO	YES	SOME	-
Q. EXPERIMENTAL MODAL ANALYSIS CAPABILITY	NO	NO	FEW	MOST	-
R. SYNCHRONOUS TIME AVERAGING (STA)	NO	NO	SOME	YES	-
S. WATERFALL PLOTTING CAPABILITY	NO	NO	FEW	MOST	-
T. RELATIVE COSTS Low End High End Nominal Expected Cost	\$400 \$2,800 \$1,200	\$1,000 \$9,000 \$6,000	\$3,500 \$18,000 \$9,000	\$4,000 \$40,000 \$15,000	\$5,000 \$100,000 \$15,000

ABBREVIATIONS

A - ACCELEROMETER
AM - ANALOG METER
FS - FREQUENCY SPECTRA
HC ONLY - PRINTED HARD COPY ONLY
LCD - LIQUID CRYSTAL DISPLAY
LT - LIVE TIME (TYPICALLY UPDATED EVERY 1-4 SECONDS)
V - VELOCITY TRANSDUCER

MS - MONOCHROME DISPLAY
OL - OVERALL LEVEL
P - PROXIMITY PROBE
RT - REAL-TIME (HIGH SPEED SCREEN REFRESH RATES)
SF - SWEPT FILTER
TWF - TIME WAVEFORM

NOTE: THE ABOVE TABLE HAS NOT TAKEN INTO ACCOUNT ALL OF THE VARIOUS INSTRUMENTS AVAILABLE ON THE MARKET TODAY. DUE TO THE RAPID ADVANCES IN TECHNOLOGY, MANY PIECES OF EQUIPMENT CANNOT BE CLEARLY DEFINED INTO THE ABOVE CATEGORIES. THIS TABLE IS ONLY MEANT TO BE USED AS AN EXAMPLE OF THE CAPABILITIES A USER WOULD EXPECT TO FIND FROM A PARTICULAR INSTRUMENT TYPE. THE ABOVE CHARACTERISTICS SHOULD BE DISCUSSED IN DETAIL WITH A PARTICULAR MANUFACTURER BEFORE PURCHASING.

NOTE: "CHARACTERISTIC" IS DEFINED FOR EACH LISTED ITEM IN SECTION 3.1. DISCUSSIONS COVERING THE GENERAL CAPABILITIES AND LIMITATIONS OF EACH INSTRUMENT TYPE FOLLOW IN SECTION 3.2.

2.23 GENERAL CAPABILITIES OF EACH VIBRATION INSTRUMENT TYPE

The following is a summary of the instruments covered previously in Table II outlining their major advantages and drawbacks.

2.231 Overall Level Vibration Meters

As the name implies, these instruments measure overall vibration (and some meters likewise measure Spike Energy, or another of the so-called ultrasonic bandpass parameters). Overall vibration refers to the overall or total amplitude summation of all the vibration in the form of acceleration, velocity, displacement, or one of the high frequency bandpass filtered parameters. At one time, these lightweight, portable instruments were used extensively, but (because of their limitations) have been replaced today by FFT Programmable Data Collectors. Some of the major drawbacks in using these instruments is their inability to display or store either spectra or time waveforms; their limited frequency ranges in most cases; and the requirement by most such meters that the vibration reading must be manually recorded which is cumbersome and time-consuming.

2.2311 Drawbacks in Measuring Only Total or Overall Vibration

For precision machinery, or for machinery that is critical to a plant's operation, routine overall vibration level recording is not sufficient. Numerous failures can occur with only a minute increase (or decrease) in the overall level, particularly if only displacement, velocity, or acceleration is used. This can occur if the problem is bearing wear, gear wear, cracked gear teeth, cracked rotor bars, etc. Even if a user feels that he can evaluate machine health by monitoring bearing condition using overall spike energy, HFD, or shock pulse, he should be aware that rolling element bearing condition is not the only cause for a high reading. Lubrication, cavitation, high pressure steam or air, gear condition, rotor rub, and belt squeal also can cause an increase in these high frequency bandpass parameters. To determine *what* is causing this parameter to increase will require a spectrum analysis or a high frequency enveloped spectrum measurement.

2.232 Swept-Filter Analyzers

These analyzers use a constant percentage analog filter (typically 2% - 10%) to sweep through a frequency range (typically from about 60 - 600,000 CPM, or 1 - 10,000 Hz). Thus, if using a 10% filter and measuring at 1000 CPM, the filter would include vibration from 950 to 1050 CPM. On the other hand, when up at a frequency of 100,000 CPM, the filter would include 95,000 to 105,000 CPM vibration. These analyzers have been replaced for the most part by FFT Programmable Data Collectors because these data collectors are capable of storing data and producing better resolved frequency spectra. However, swept-filter analyzers can be still used for field balancing, strobe light "slow motion studies" and phase analysis. A drawback is that the operator has to be near the instrument to use the strobe and tune in to the frequency. Some other drawbacks are they are not easily transported since they are fairly large and heavy in size and are too cumbersome to be used on a predictive maintenance route. Besides, the data cannot be stored by these analyzers, nor are they capable of storing or displaying time waveforms.

The operator has to be aware that swept-filter analyzers only capture events which occur when the filter happens to be measuring at the event's (i.e., transient vibration spike) frequency setting at a particular instant in time. The frequency precision is limited by the filters used. In all cases, the frequency resolution is limited to the filters used.

2.233 FFT Programmable Data Collectors

FFT Programmable Data Collectors are the current state-of-the-art instruments of choice for predictive maintenance programs. The FFT capability transforms the time waveforms captured by these units into frequency spectra and most data collectors can display them on a small LCD screen in "live-time". Although the data collector was designed to collect data on many pieces of machinery, many of them can also be used as an analyzer in the field due to their graphics, FFT, live-time capabilities, and their ability to display time waveforms. Most of them can measure phase with the attachment of a strobe light or phototach and can also measure high frequency parameters such as HFD and Spike Energy. Some can also measure high frequency enveloped spectra, such as "Spike Energy spectra"¹¹, "Amplitude Demodulated spectra"¹², "Acceleration Enveloped spectra"¹³ or "SEE spectra"¹⁴.

While most of the data collectors are single channel instruments, others have anywhere from 2 to 4 channels. Only a few FFT data collectors are capable of multi-channel data input which is needed for modal analysis and helpful in operating deflection shape tests.

Some data collectors have the ability to measure 3 different parameters (i.e., velocity, acceleration, and spike energy) simultaneously with one push of the "store" button. Also, if a triaxial accelerometer is used, multi-channel units having at least 3 channels can display the spectra from all 3 directions simultaneously, with little or no loss in analyzer processing speed.

The frequency range of the average data collector is normally from 60 to 1,500,000 CPM (1 to 25,000 Hz), but some data collectors are now available with frequency measuring capabilities as low as 6 CPM (0.1 Hz), and as high as 360,000 CPM (6000 Hz). With the intense development of data collectors going on today, no telling what their capabilities might be only within the next 5 to 10 years.

2.234 Real-Time Spectrum Analyzers

The real-time spectrum analyzer is the most powerful diagnostic tool for advanced diagnostic techniques on the market. The "real-time" display updates "quicker than the eye" when the frequency span and other setup parameters are properly specified, as opposed to the "live-time" display in data collectors. In addition many of them have a built-in time buffer which allows one to store "runups" or "coastdowns" and play them back over and over again (similar to using a tape recorder). They also can capture short duration transient (less than 20 milliseconds) events and examine the data looking for potential problems.

Real-time analyzers are excellent in performing impulse natural frequency tests, coastdown/runup tests, and transient capture due to their "peak hold" capabilities. They can also generate Bode' and Polar plots to verify the location of natural frequencies. The multi-channel capabilities available in many of these units provide an excellent facility to capture data in operating deflection shape and modal analysis. Most are capable of performing synchronous time averaging and order tracking. Phototach input is available for phase analysis (or multi-channel RTA's can use another accelerometer or force transducer as a reference for phase measurement).

Since a real-time spectrum analyzer is normally complex, the user will require additional training and frequent use to remain proficient. Also, it is not usually very portable. Some are now equipped with a 3.5 inch (1.44 Mb) floppy drive (or even large megabyte hard drives) which provide a virtually limitless storage capacity for spectral and time waveform data. Recent RTA's have built-in computers with special cards which allow PMP software to be installed on them. Some are also equipped with word processing, spreadsheets, and graphics software to provide "on-the-spot" report generation.

The typical real-time analyzer today is capable of measuring frequencies in a very wide range from 0 - 6,000,000 CPM (0 - 100,000 Hz). In summary, the real-time analyzer can prove invaluable for experienced analysts who want to conduct sophisticated diagnostic investigations. However, with the constantly increasing capabilities of today's data collectors, they are no longer an absolute necessity for a "complete" condition monitoring program.

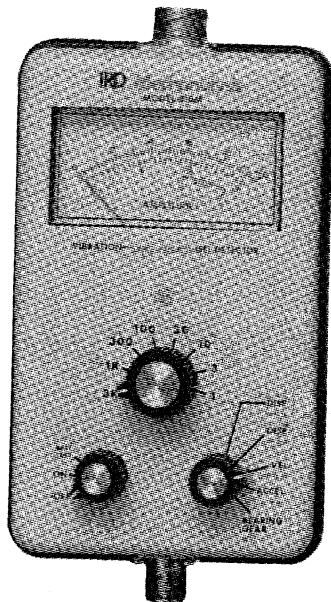
2.235 Instrument Quality Tape Recorders

Instrument quality tape recorders can simultaneously record many different signals (whether from vibration pickups, pressure transducers, tachometers, current transformers, phototachs, etc.). Tape recorders can capture short-lived transient events which cannot even be "seen" by an analyzer. After a signal has been captured on tape, it can be analyzed back in the office at a much lower speed to allow capture of short-lived transients, particularly since it may be played back over and over again.

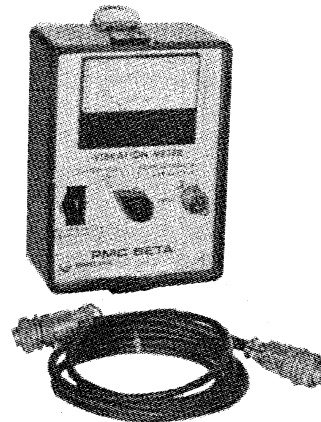
There are two types of tape recorders available today: analog and digital. The analog type records the actual signal input from the transducer without breaking it into a number of sampled points. However, their dynamic range is limited to 40 - 48 dB. On the other hand, digital tape recorders typically have a dynamic range of about 72 - 80 dB. Therefore, they are about 30 dB more amplitude sensitive. That is, digital recorders can detect small amplitude frequencies with amplitudes over 30 times lower than can analog devices in the simultaneous presence of much higher amplitude frequencies. Incidentally, the digital tape recorder samples the input signal at a specified rate and reproduces it as stored numbers.

Tape recorders accurately record the time waveform of the vibration which can be analyzed later with a real-time spectrum analyzer or data collector. Both types of tape recorders have multi-channel capabilities (up to 64 channels or more) which allow the capture of many data points simultaneously.

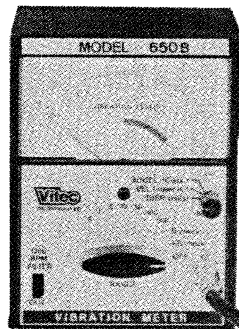
However, instrument quality tape recorders can be very costly. As an alternative, they can be rented for use. The price then will depend on the frequency range, the number of channels required, and the type. Furthermore, they may be somewhat complicated to use, so frequent use to remain proficient may be required.



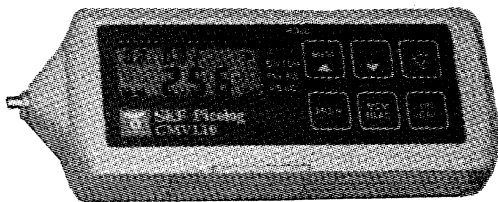
A. IRD 810 VIBRATION SPIKE
ENERGY DETECTOR



B. PMC/BETA 201A
VIBRATION METER



C. VITEC 650
VIBRATION METER

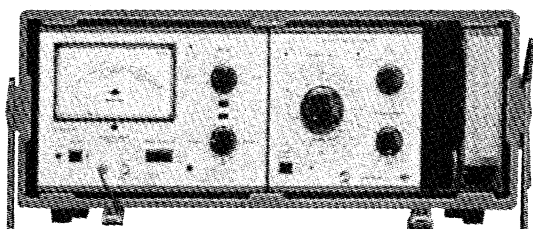


D. SKF CMVLIO PICOLOG
VIBRATION / SEE / ACCELERATION
ENVELOPE DETECTOR

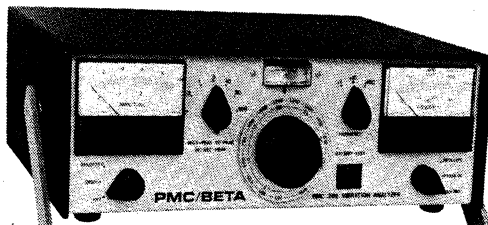


E. SKF CMVP20
VIBRATION PEN

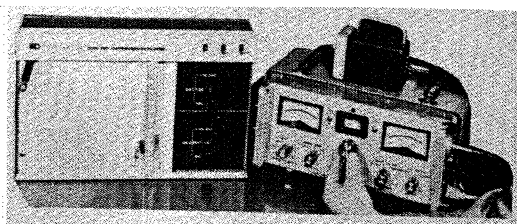
ILLUSTRATION A EXAMPLES OF VARIOUS HAND-HELD OVERALL VIBRATION METERS



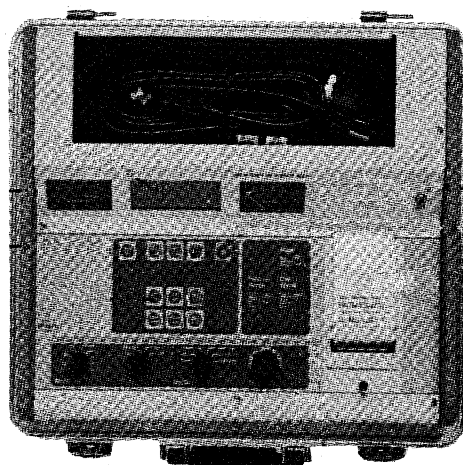
A. B&K 3513



B. PMC/BETA 208



C. IRD 350 WITH
IRD 1080 PLOTTER



D. IRD 880

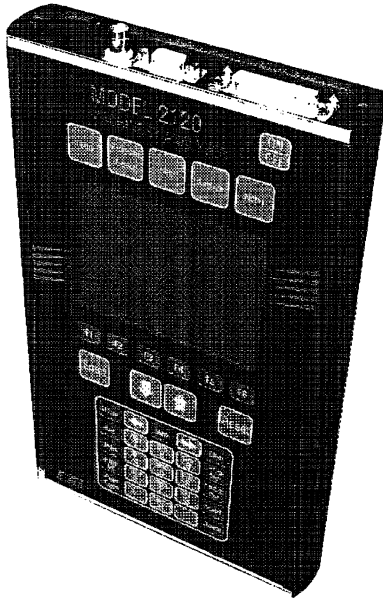


E. SCHENCK VIBRATION 30

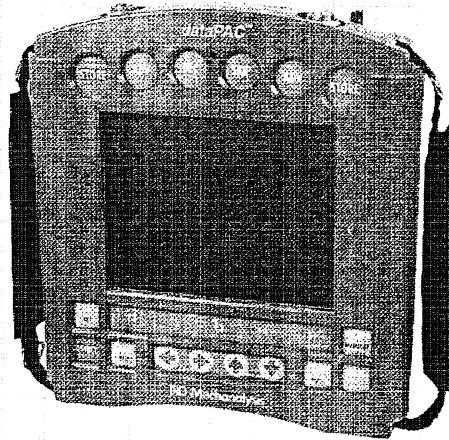


F. VITEC VIBRA/VIEW

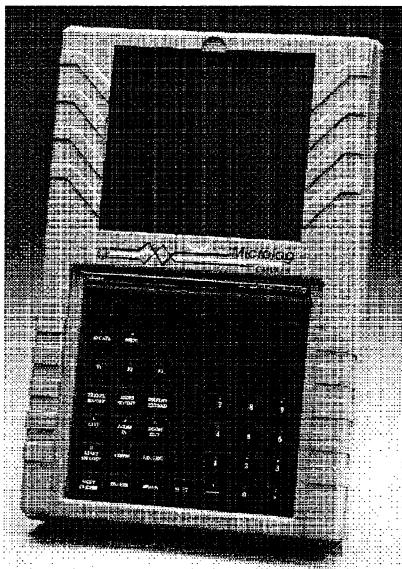
ILLUSTRATION B EXAMPLES OF PORTABLE SWEPT-FILTER ANALYZERS



**A. CSI 2120
MACHINERY ANALYZER**



ENTEK IRD DataPac 1500



C. SKF MICROLOG CMVA55

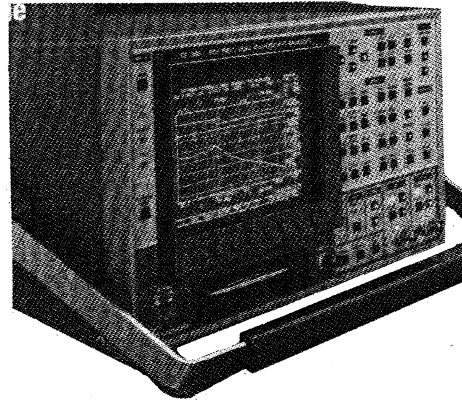


**D. PREDICT/DLI WATCHMAN DC-7B
(3 CHANNEL)**

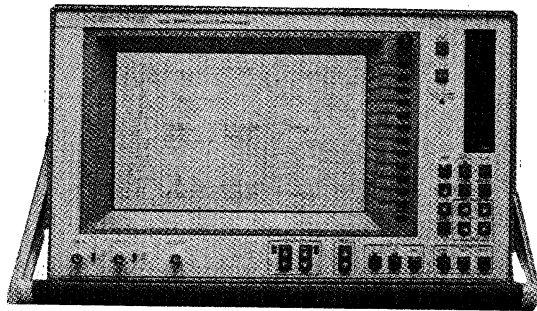
**ILLUSTRATION C
EXAMPLES OF VARIOUS DATA COLLECTORS**



**A. HEWLETT PACKARD 35670A
DYNAMIC SIGNAL ANALYZER
(2 OR 4 CHANNEL)**



**B. ONO SOKKI CF-360
(2 CHANNEL)**



**C. ROCKLAND 90665L
DYNAMIC SIGNAL ANALYZER
(2 CHANNEL)**

**D. SIGNAL PROCESSING SYSTEMS SA 390
DYNAMIC SIGNAL ANALYZER
(2, 4, 6, OR 8 CHANNEL)**

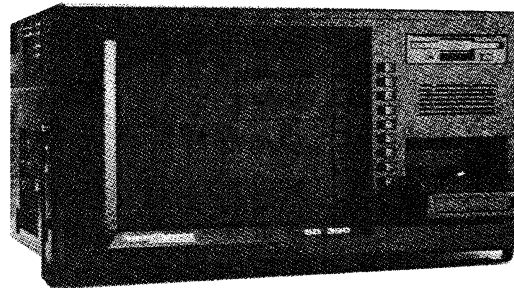
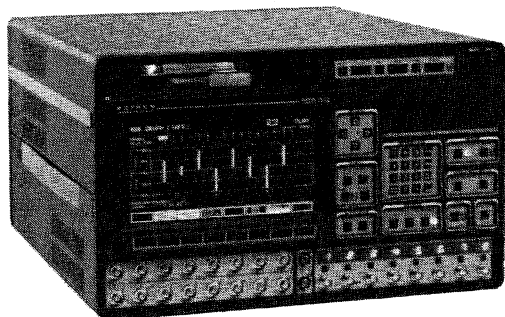
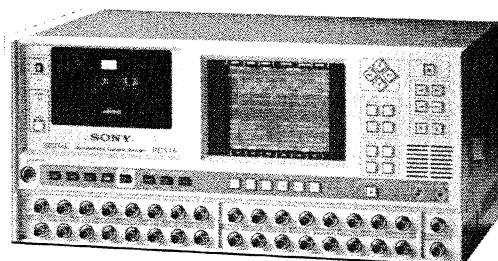


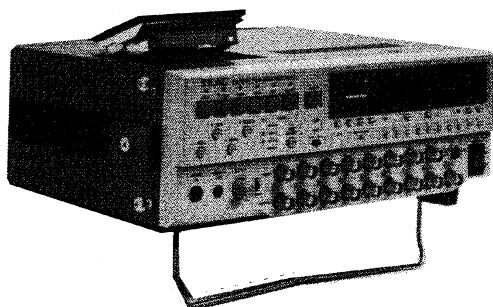
ILLUSTRATION D EXAMPLES OF VARIOUS REAL-TIME ANALYZERS



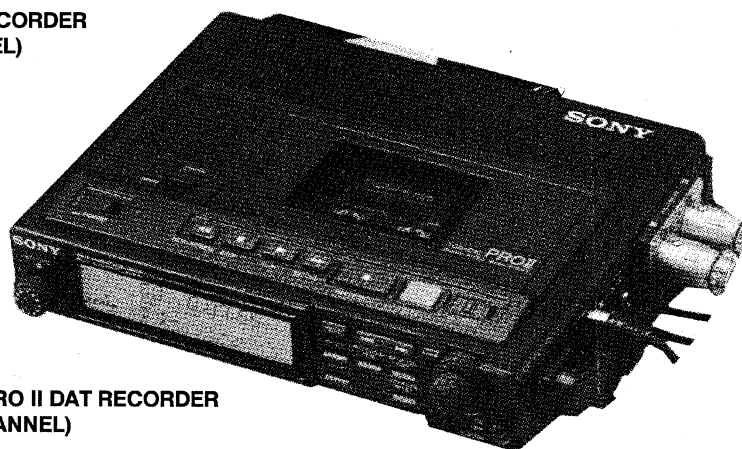
**A. METRUM RSR 512 DAT RECORDER
(8,16,24 OR 32 CHANNEL)**



**C. SONY PC 116 DAT RECORDER
(2 THRU 16 CHANNEL)**



**B. TEAC RD-130T DAT RECORDER
(2, 4, OR 8 CHANNEL)**



**D. SONY TCD-D10 PRO II DAT RECORDER
(2 CHANNEL)**

ILLUSTRATION E EXAMPLES OF INSTRUMENT QUALITY TYPE RECORDERS

2.3 OVERVIEW OF VIBRATION TRANSDUCERS AND HOW TO PROPERLY SELECT THEM

2.3.1 INTRODUCTION

Each of the following transducers will be covered:

- 1) Accelerometers
- 2) Velocity Pickups
- 3) Non-contact Eddy Current Displacement Probes
- 4) Shaft Contact Displacement Probes (including Shaft Sticks and Shaft Riders)

Figure 22 includes the three most common transducers in use today which include the accelerometer, velocity pickup and non-contact displacement probes. Table III is a general summary of the various categories of accelerometers, velocity pickups, and non-contact probes showing the more important characteristics and general specifications. In addition, model numbers of such transducers made by various manufacturers are included as examples.

In this chapter the optimum applications of each of these transducers (depending on the measurement to be made), mounting techniques and their influence on the accuracy of the vibration measurement will be discussed.

It is important to note that the following information is meant to be a general overview of transducer design and utilization. The specific design characteristics for each transducer may vary individually from vendor to vendor. Therefore, the reader should use this chapter as a guide for discussing his needs with a qualified vendor before purchasing a particular transducer or set of transducers.

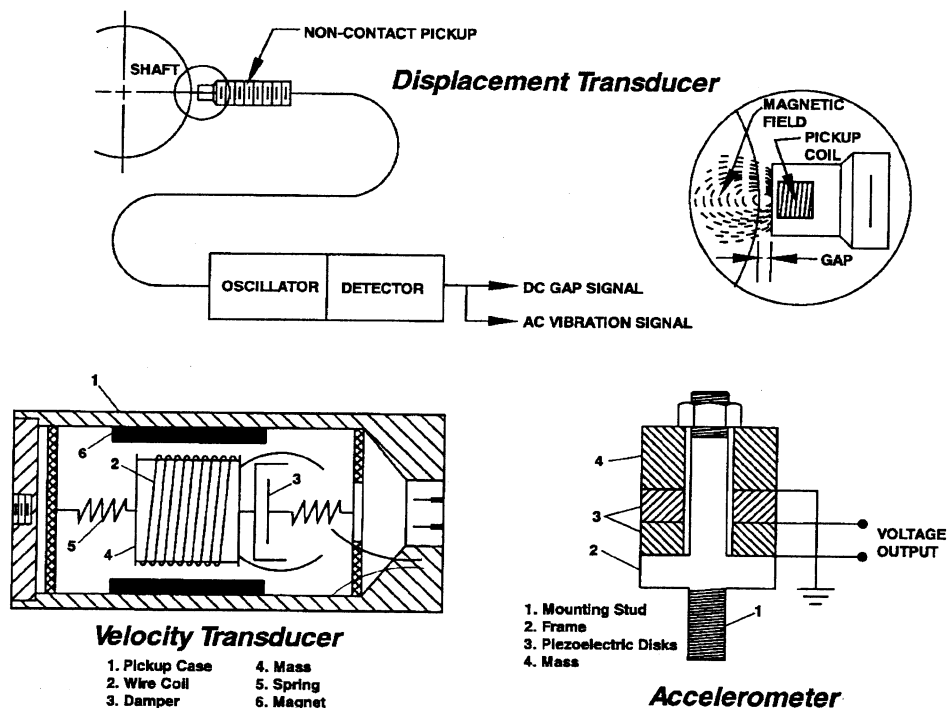


FIGURE 22
EXAMPLES OF VARIOUS TYPES OF VIBRATION TRANSDUCERS

2.32 TYPES OF VIBRATION TRANSDUCERS AND THEIR OPTIMUM APPLICATIONS

It is important that a vibration analyst understand the limitations and optimum applications of each of these four types so that he can use them to his best advantage. At times, each type has certain characteristics that justifies its use in a particular application over the other types when monitoring machinery. Accelerometers and velocity pickups placed on bearing housings measure absolute vibration while non-contact displacement probes measure relative vibration.

2.321 Accelerometers

As the name implies, accelerometers are sensors which provide the direct measurement of acceleration (g). It is the piezoelectric element in the accelerometers that produces a signal proportional to acceleration. Accelerometers measure absolute vibration.

Accelerometers are the most common PMP transducers in use today due to 3 primary reasons: (1) they are relatively inexpensive when compared to velocity pickups; (2) their frequency range capabilities are much wider than those of velocity pickups, non-contact probes, shaft sticks, or shaft riders; and (3) much more funding is being expended in research and development of a large variety of accelerometers, not only to lower their cost, but also to enable them to accurately measure both lower and higher frequency data and to withstand harsh environmental conditions (high temperatures, operation in submerged oil baths or under water, exposure to corrosive gases or liquids, etc.).

There are three primary performance characteristics of accelerometers which affect their performance. These include voltage sensitivity (mV/g), frequency response (Hz or CPM), and mass (grams). In order to determine the acceleration in g's from the voltage generated by the piezoelectric crystal in the accelerometer, consider the following example:

EXAMPLE: If 2 Volts are generated by an accelerometer, which has a sensitivity of 100 mv/g, the acceleration would be:

$$A = \frac{2 \text{ Volts}}{.1 \text{ Volt/g}} = \underline{\underline{20g}}$$

In most cases, if desiring to take low frequency measurements, it will be necessary to choose one of the low frequency accelerometers listed in Table III which typically are much higher in weight and normally have voltage sensitivities (mV/g) much higher than the general purpose accelerometers. The reason for this high voltage sensitivity is to bring the vibration signal above the noise. Even though the displacement (mils) levels may be high, the acceleration levels (g) will be low for vibration at frequencies of less than 60 CPM (1 Hz). On the other hand, if desiring to take high frequency measurements (typically above 600,000 CPM or 10,000 Hz), it will likely be necessary to acquire one of the high frequency accelerometers listed in Table III. Typically, these high frequency accelerometers are much smaller with a lower voltage sensitivity, usually on the order of 10 mV/g or less.

One of the common misconceptions is that the higher the voltage sensitivity (mV/g), the better the accelerometer. This is not always the case. For example, some seismic accelerometers used in low frequency measurements are very sensitive to temperature changes and, if dropped, can fail due to the instantaneous voltage surge which oversaturates their built-in electronics. Conversely, a general purpose or high frequency accelerometer can be dropped with no damage to the transducer.

**TABLE III
GENERAL TRANSDUCER CHARACTERISTICS**

TYPICAL SPECIFICATIONS	ACCELEROMETERS					VELOCITY PICKUPS		NONCONTACT DISPLACEMENT PROBES
	GENERAL PURPOSE	LOW FREQUENCY	HIGH FREQUENCY	PERMANENTLY MOUNTED	TRIAXIAL	SEISMIC	PIEZOELECTRIC	
TYPICAL SENSITIVITY RANGE (mV/g Unless Noted)	10 - 100	500 - 10,000	0.4 - 20	10 - 100	10 - 100	(mV/in/s) 100 - 1000	(mV/in/s) 100 - 1000	(mV/mil) 100 - 200
TYPICAL MEASUREMENT FREQUENCY RANGE (CPM)	60 - 600,000	6 - 80,000	600 - 3,600,000	180 - 600,000	120 - 600,000	600 - 60,000	60 - 270,000	0 - 120,000
TYPICAL STUD MOUNTED NATURAL FREQUENCIES (CPM)	960,000 - 2,700,000	390,000 - 2,100,000	4,200,000 - 10,800,000	1,080,000 - 1,800,000	480,000 - 1,200,000	APPROXIMATELY 200,000	APPROXIMATELY 200,000	N.A.
TYPICAL WEIGHT RANGE (grams)	17 - 110	135 - 1000	1.2 - 15	60 - 180	17 - 130	100 - 150	150 - 250	230 - 280
TYPICAL USABLE TEMPERATURE RANGE (°F)	(-100°) - 250°	(-100°) - 250°	(-40°) - 350°	(-50°) - 250°	(-100°) - 250°	(-50°) - 250°	(-50°) - 250°	(-30°) - 350°
EXAMPLE TRANSDUCER MODEL NUMBERS AND MANUFACTURER (Alphabetical Order)	B & K 4370/81 B & K 4390 IRD 943 IRD 970 PCB 302A06 PCB 302A09 PCB 307A PCB 308B RION PV-44A VIBRAMET. 1020 VIBRAMET. 1021 WILCOXON 793	B & K 8318 (S) PCB 326A01 (S) PCB 326A02 (S) PCB 326A03 (S) PCB 393C (C) VIBRAMET. 5100 (S) WILCOXON 731 (C) WILCOXON 793L (C) WILCOXON 797L (S)	B & K 5309 PCB 303A02 RION PV 908 VIBRAMET. 6032 WILCOXON 732	B & K 8310/24 B & K 8317 IRD 941B IRD 942 PCB 328B VIBRAMET. 1024	B & K 4321 PCB 306A PCB 306A06 VIBRAMET. 3130 WILCOXON 733	BENTLY. 47533 IRD 544	BENTLY. 26862 IRD 560 WILCOXON 756V WILCOXON 793V WILCOXON 797V	BENTLY. 3300 BENTLY. 7200 IRD 403
		(S) - SHEAR MODE (NOT TEMPERATURE SENSITIVE; BRIEF SETTling TIME) (C) - COMPRESSION MODE (TEMPERATURE SENSITIVE; LONG SETTling TIME)						

NOTES:

1. For detailed information on particular transducers, see the APPENDIX which includes specifications for many of them.
2. By inclusion of specific transducers within this table, Technical Associates does not necessarily recommend them.

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Note that Table III also lists some accelerometers that are specially designed for permanently mounting even in harsh environments. Sometimes they are placed under water or in lubricating or cutting oils. Others are designed to make triaxial measurements (simultaneous measurements in the horizontal, vertical and axial directions) to increase the measurement speed on a PMP route.

Figures 23 thru 25 illustrate the three most commonly used piezoelectric accelerometers. There are two major types. Figures 23 and 24 show one type, called the "compression mode accelerometers". Note that the accelerometer in Figure 24 is an inverted compression type. Figure 25 shows the other major type, known as a "shear mode accelerometer". Until recently, the compression designs in Figures 23 and 24 probably have been the most widely utilized designs, mainly due to their simplicity and lower cost, along with their wide frequency range.

However, one of the disadvantages of compression mode accelerometers is that they are often adversely affected by thermal transients and base strain sensitivity which can oversaturate their electronics. The time it will require for the transducer to "settle" will be much longer than that expended for a shear mode accelerometer (discussed below), particularly when making low frequency measurements. In those situations where an accelerometer will be subjected to large differences in temperature (either due to a temperature change in the mounting surface or high pressure air blowing continuously on the accelerometer), the shear type accelerometer pictured in Figure 25 may be a better choice since the crystal element is isolated from the base and housing by being sandwiched between the seismic mass and a center post. The

advantages of shear mode accelerometers are a stable output signal (especially when measuring at low frequencies) and smaller size and mass. The disadvantage of these units is the higher cost (in most cases) due to the added components required to make up the shear configuration.

This difference between shear mode and compression mode accelerometers is particularly evident when an analyst is taking low frequency measurements (particularly below 60 CPM or 1 Hz). In this case, a compression mode accelerometer may require 3 or 4 minutes to stabilize before any measurements can be taken whatsoever. However, if a shear mode accelerometer was used instead, it will not be nearly so sensitive to thermal transients. It will stabilize quickly, allowing the analyst to begin his measurement almost immediately, or certainly within 20 seconds or so.

Another important difference between piezoelectric accelerometers are the two types of signal conditioning electronics that they can have. Each of these is pictured in Figure 26: (1) A high impedance, charge mode type requires an external signal conditioner; and (2) a low impedance, voltage mode, "ICP" type contains built-in signal conditioning electronics and does not normally require any external signal conditioning (meaning that the accelerometer can be powered directly from ICP¹⁵ circuitry built right into most data collectors and real time analyzers today. Each of the three accelerometers pictured in Figures 23 thru 25 are the "ICP" (integrated circuit piezoelectric) type - the most commonly used accelerometer types in programs today.

In the case of the charge mode unit, its accelerometer sensitivity is usually defined in units of picocoulombs per g (pC/g), whereas the sensitivity of voltage-mode units is normally expressed directly in millivolts per g (mV/g). These charge mode accelerometers will most always require an external charge amplifier/signal conditioner in the field to use them.

The built-in microelectronics conditions the signal from the crystal within the ICP accelerometer to a low impedance voltage which is compatible with the readout device. To turn on and perform its measurement, a constant current power source (normally available in most all FFT analyzers today) is required for the transistor within the accelerometer. The advantages of the ICP accelerometers are: (1) their fixed sensitivity; (2) the ability to operate in dirty and moist environments reliably; (3) they only need an ordinary 2-wire coaxial cable which can be usually powered without additional power sources up to about 1000 feet; (4) they do not need a separate signal conditioner when one is available in the analyzer; and (5) the system usually costs less. One disadvantage is a limited temperature range due to the survivability of the built-in electronics (although this can sometimes be overcome by use of special ceramic construction materials). Another disadvantage occurs if the accelerometer is not hermetically sealed - its low frequency performance will be degraded significantly if moisture enters into the transducer. Reference 4 points out that "all accelerometers breathe, unless they are certified as hermetically sealed, and during this micro-breathing, moisture inevitably penetrates." Therefore, it is important to acquire a hermetically sealed unit, particularly if it is to be used in humid environments, and especially in those cases where low-frequency measurements are to be conducted.

On the other hand, one advantage of the charge mode accelerometer is the ability to operate at high temperatures. However, one of the biggest disadvantages of the charge mode accelerometer is that a separate charge amplifier is required between the analyzer and the sensor, as shown in Figure 26. Because of this disadvantage and other disadvantages, such as requiring a special low-noise, fixed length cable and a clean and moisture-free environment, they are not used in most predictive maintenance measurement systems today.

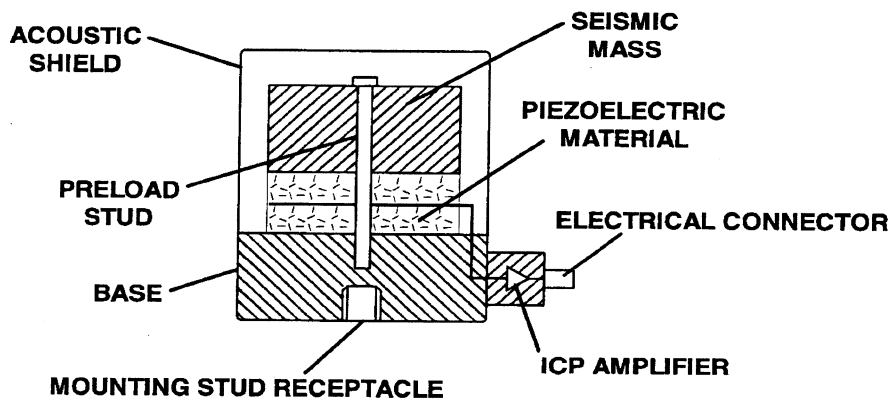


FIGURE 23
COMPRESSION TYPE ACCELEROMETER (Ref. 3)

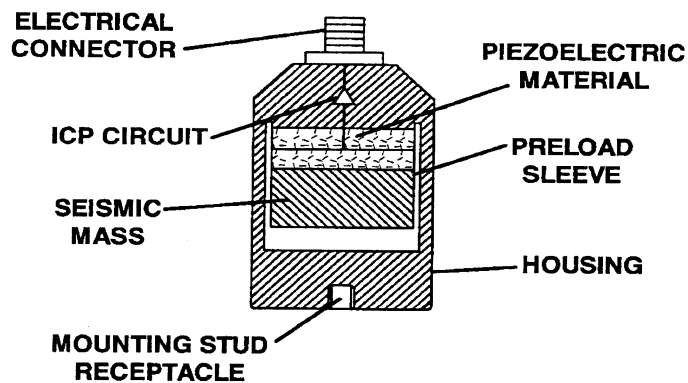


FIGURE 24
INVERTED COMPRESSION TYPE ACCELEROMETER (Ref. 3)

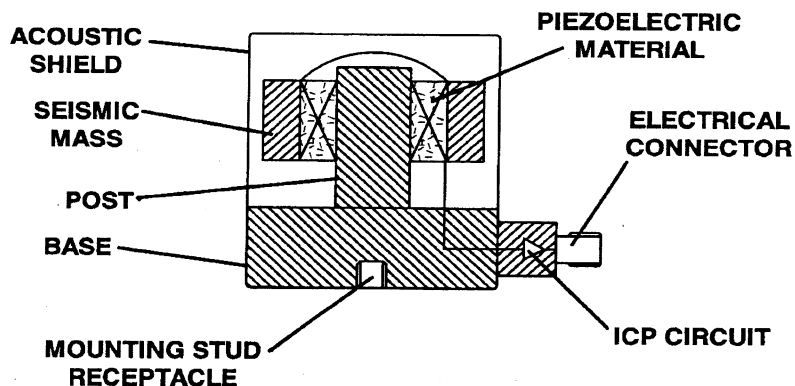


FIGURE 25
SHEAR TYPE ACCELEROMETER (Ref. 3)

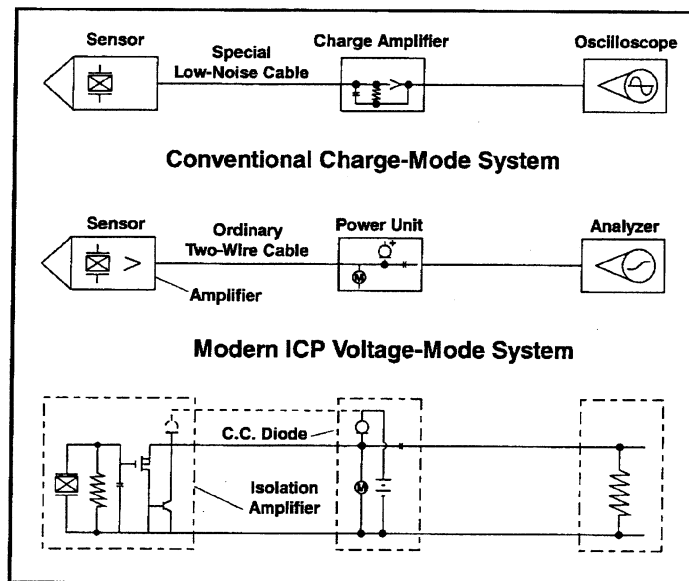


FIGURE 26
COMPARISON OF CHARGE MODE AND ICP VOLTAGE-MODE
ACCELEROMETERS (Ref. 4)

When choosing an accelerometer, it is important to be aware of the resonant frequency and phase response of each accelerometer. This is illustrated by Figure 27 which shows an accelerometer with a fairly constant phase response from approximately 5% up to about 60% of its mounted natural frequency (f_m). Note that as the mounted natural frequency of an accelerometer is approached, a dramatic phase change takes place. The phase shifts almost 180° after passing through resonance. Therefore, when an accelerometer is used to perform a phase analysis on a machine, the frequency range at which this phase response occurs must be taken into consideration, particularly on very high speed machinery. If not, the accuracy of the phase analysis could be adversely effected. Furthermore, if a magnet or an extension probe is used to mount an accelerometer, the natural frequency of the system will be considerably lower than the accelerometer stud mounted. As a result, the frequency ranges from which accurate phase measurements can be obtained will be considerably lower.

Accelerometers are available with widely different frequency responses. For Example, frequency ranges go from as low as .01 Hz (.6 CPM) up to over 60,000 Hz (3,600,000 CPM); voltage sensitivities will vary from about 0.4 mV/g up to 10,000 mV/g (10 V/g); accelerometer masses can vary from 1 gram up to over 1000 grams. Since most predictive maintenance programs include not only normal speed machinery (600 to 3600 RPM), but also low-speed (particularly below 200 RPM), as well as high-speed machinery (particularly above 10,000 RPM), it will be necessary (in most cases) to have at least three different accelerometers - a low frequency, a general purpose, and a high frequency accelerometer. Most likely, the general purpose accelerometer can be used successfully to make over 90% of the measurements. However, without the low and high frequency accelerometers, the accuracy of the measurements from the very low or very high speed machines will be reduced. Therefore, the data integrity will be compromised.

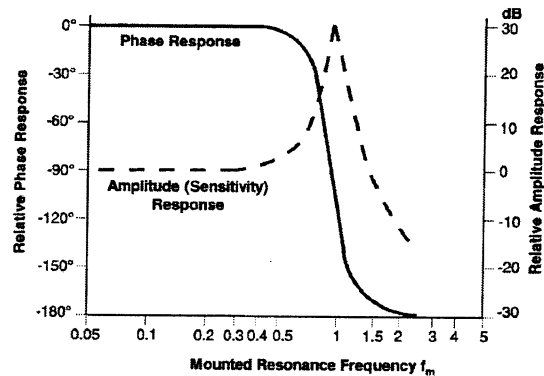


FIGURE 27
BODE' PLOT OF ACCELEROMETER SENSITIVITY AND PHASE RESPONSE
AS A FUNCTION OF FREQUENCY (Ref. 4)

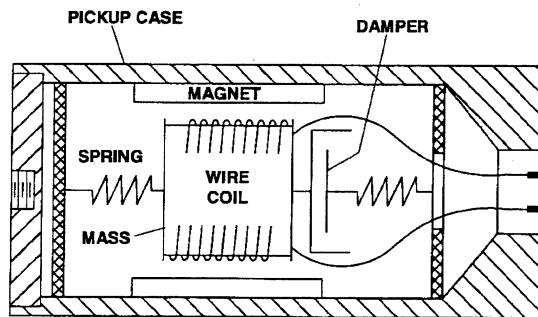
2.322 Velocity Pickups

For most applications today, velocity pickups have been replaced by accelerometers because of their limited frequency ranges and relatively higher cost. Also, these units are typically very large in size and weight (the example in Figure 28 weighs 21 ounces or 595 grams), which can undesirably contribute to the mass load of the machine or structure being evaluated. However, one advantage of a velocity pickup is that it does not have to be powered by an external power supply (such as an ICP current or a separate charge amplifier). Also, since velocity is the commonly preferred unit of vibration measurement in most PMP programs, the signal does not have to be integrated from acceleration to velocity (as is the case with accelerometers). Therefore, the integration noise which often appears in the first 2 to 4 lines of the FFT spectra taken by accelerometers integrated to velocity does not occur with these pickups (however, the velocity pickups can still produce noise within the first few lines of the spectrum due to their electronics, the insufficient analyzer input sensitivity and depending on the frequency span chosen).

There are two types of velocity transducers - moving coil seismic velocity transducers and piezoelectric velocity pickups. A seismic velocity transducer is shown in Figure 28 while example piezoelectric velocity pickups are shown in Figures 29 and 30. Velocity transducers also measure absolute vibration, like accelerometers.

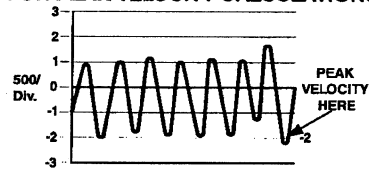
Unlike piezoelectric transducers (piezoelectric accelerometers and piezo-velocity), the moving coil seismic velocity pickup does not require an external power supply of any kind. The voltage is self-generated. The seismic mass and coil remain stationary since they are spring-suspended, whereas the permanent magnet surrounding the coil will oscillate since it is firmly attached to the transducer case. A voltage (proportional to the relative velocity of the magnet and coil) is generated and then transmitted from the transducer into a vibration analyzer.

Since its seismic mass is supported by soft springs, this transducer has a very low natural frequency of about 10 - 20 Hz (600 - 1200 CPM). The suspended mass shown on Figure 28 is damped (either electrically or with synthetic oil). Therefore, the transducer's low frequency response is limited by its first natural frequency. As the mass overcomes the damping effect in low frequencies and begins to move in phase with the vibration, its sensitivity will drop because the coil will hardly be cutting through the magnetic flux. For example, if an IRD 544 velocity pickup like that shown in Figure 28 was used to measure vibration below 10 Hz (600 CPM), a Correction Factor Chart, as shown in Figure 29, would have to be used. If a measurement of .1 in/sec at 300 CPM were recorded, Figure 29 shows that the actual level



Frequency Range ($\pm 10\%$): 600-60,000 CPM (10-1000 Hz)
 Sensitivity ($\pm 10\%$): 764 mV RMS / in/sec PK
 = 1080 mV Peak / in/sec PK
 Temperature Range: -40° F to + 500° F
 (-40° C to + 260° C)
 Impedance: R = 2000 ohms; L = 0.7 Hy
 Damping Factor: 0.7 (critical)
 Transverse Response (ave): Less than 5% (10-1000 Hz)
 Magnetic Field Sensitivity: 0.15 in/sec / Gauss
 Weight: 21oz (595 grams)

FROM SCOPE DISPLAYS FOR PEAK VELOCITY CALCULATIONS



$$\begin{aligned} \text{PEAK VEL.} &= 2 \times 500 \text{ MV} = 1000 \text{ MV} \\ \text{VEL. PK} &= \frac{1000 \text{ MV}}{1080 \text{ MV PEAK/in/sec. PK}} \\ &= .926 \text{ in/sec. PK} \end{aligned}$$

FIGURE 28
IRD 544 SEISMIC VELOCITY PICKUP (Ref. 5)

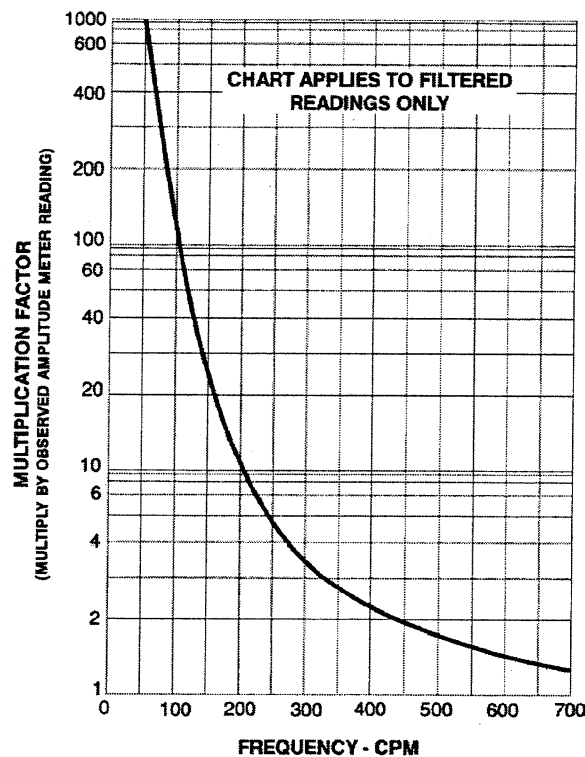


FIGURE 29
CORRECTION FACTOR CHART FOR IRD 544 SEISMIC VELOCITY PICKUP
WHEN MEASURING FREQUENCIES BELOW 600 CPM (10 Hz) (Ref. 5)

would be about 3.3 times higher (or about .33 in/sec). Note that with decreasing frequencies, significantly higher and higher multiplication factors must be applied to correct the vibration readings. Therefore, a seismic velocity pickup is not a good low frequency transducer.

Another problem with most seismic velocity pickups is the upper frequency response which is commonly limited to no more than about 1000 - 2000 Hz (60,000 to 120,000 CPM). If a much lower or higher frequency measurement is required, a piezoelectric velocity transducer, as shown in Figures 30 or 31, could be used, particularly if a direct velocity reading is preferred. The direct velocity reading is provided by an internal analog integrator inside the transducer which converts the acceleration signal to velocity. The noise which generally appears at low frequencies due to integration and other sources is suppressed by having this analog integration performed within the transducer rather than by an integration circuit in the instrument. In comparison to seismic velocity transducers, the frequency range of piezoelectric velocity pickups is much wider. For example, note that the unit in Figure 30 has a ($\pm 10\%$) frequency response of 60 - 270,000 CPM (1 - 4500 Hz) and that of Figure 31 is from 150 - 210,000 CPM (2.5 - 3500 Hz); while the seismic unit in Figure 28 has a ($\pm 10\%$) frequency response of only 600 - 60,000 CPM (10 - 1000 Hz).

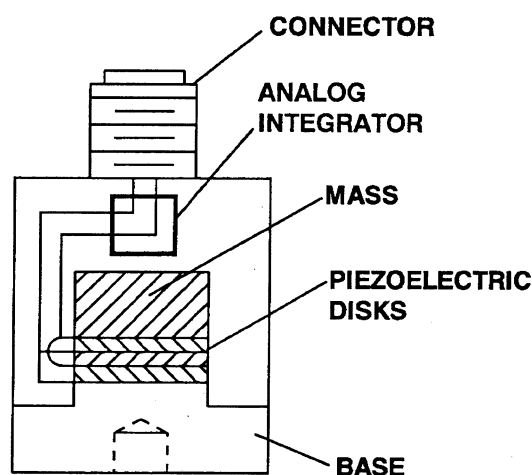


FIGURE 30
IRD 560 PIEZOELECTRIC VELOCITY PICKUP (Ref. 5)

Frequency Range ($\pm 10\%$):	60-270,000 CPM (1-4500 Hz)
Sensitivity ($\pm 10\%$):	764 mV RMS / in/sec PK = 1080 mV Peak / in/sec PK
Temperature Range:	-40° F to + 250° F (-40° C to + 121° C)
Mounted Natural Freq.:	600,000 CPM min. (10,000 Hz)
Transverse Response :	5% max @ 600 CPM (10Hz), 5 g peak
Magnetic Field Sensitivity:	.00001 in/sec /Gauss
Weight:	38 oz (1079 grams)

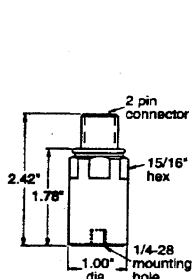
Figure 30 shows a piezoelectric disk mounted below the seismic mass. Be aware that this disk introduces only a small amount of phase shift. This makes a piezoelectric velocity pickup a much better choice for phase analysis and balancing than a seismic velocity pickup. In addition, unlike seismic transducers, piezoelectric velocity pickups and accelerometers are virtually unaffected by magnetic fields (see comparative magnetic field sensitivities in Figures 28 and 30, respectively), making them ideal for analyzing electric motors.



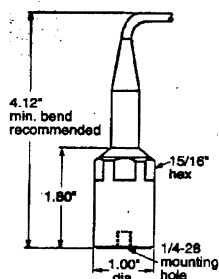
793V Series Piezoelectric Velocity Transducers

FEATURES:

- Intrinsic safety certified option
- Industrial ruggedness
- Eliminates distortion caused by high frequency signals
- Corrosion-resistant
- Internally integrated to velocity
- Ultra low-noise electronics for clear signals at very low vibration levels
- Miswiring protection



**793V Series
Models**



**793VS Series
Models**

SPECIFICATIONS

DYNAMIC

	All Models	793V	793V-5	Units
Sensitivity, $\pm 10\%$, 25°C	—	100	500	mV/in/sec
Velocity Range	—	50	10	in/sec peak
Amplitude Nonlinearity	1%			
Frequency Response:				
$\pm 10\%$	—	2.0 - 3,500	8.0 - 3,500	Hz
$\pm 3\text{ dB}$	—	1.5 - 7,000	5.0 - 7,000	Hz
Resonance Frequency, mounted, nom.	15 kHz			
Transverse Sensitivity, max.	5 % of axial			
Temperature Response	see graph			

ELECTRICAL

Power Requirement:				
voltage source	18 - 30 VDC			
current regulating diode ¹	2 - 10 mA			
Electrical Noise, equiv. in/sec, nominal:				
Broadband 2.5 Hz to 25 kHz	—	100	40	$\mu\text{in/sec}$
Spectral 10 Hz	—	10	4.0	$\mu\text{in/sec}/\sqrt{\text{Hz}}$
100 Hz	—	1.0	0.4	$\mu\text{in/sec}/\sqrt{\text{Hz}}$
1000 Hz	—	0.2	0.05	$\mu\text{in/sec}/\sqrt{\text{Hz}}$
Absolute Phase Shift, nom. the greater of:	—	$\tan^{-1} 2f$ or 2°	$\tan^{-1} 5f$ or 2°	
Output Impedance, nominal 4mA supply, the greater of:	—	5,000 Ω or 200 Ω	25,000 Ω or 200 Ω	
Bias Output Voltage, nominal	10 VDC			
Grounding	case isolated, internally shielded			

ENVIRONMENTAL

Temperature Range	-50 to 120°C
Vibration Limit	250 g peak
Shock Limit	2,500 g peak
Electromagnetic Sensitivity, equiv. in/sec	25 $\mu\text{in/sec/gauss}$
Sealing	hermetic
Base Strain Sensitivity	0.0005 in/sec/ μstrain

PHYSICAL

Weight	145 grams
Case Material	316L stainless steel
Mounting	$\frac{1}{4}$ - 28 tapped hole
Output Connector	MIL-C-5015 2-pin
Connections	Pin A signal, power Pin B common
Cabling ² : Mating Connector	Wilcoxon R6 (Amphenol 97-3106A-10SL-4S)
Recommended Cable	J9T2, two conductor shielded, Tefzel jacket, 30 pF/ft.

NOTES: ¹ A maximum current of 6 mA is recommended for operating temperatures in excess of 100°C .

² Cabling for "S" option model is J61, coaxial, polyurethane jacket, 30 pF/ft.

OPTIONS: Intrinsic safety certification; stainless steel flexible hose; 10 ft. sealed integral cable "S" or "SG", temperature sensor "T", customer specified sensitivity, low frequency filtering, cable length, connector.

ACCESSORIES SUPPLIED: SF6 mounting stud, calibration data, CCD card

ACCESSORIES AVAILABLE: R6SL-J9T2-XX splash-proof cable assembly, magnetic mounting bases, SF8 cementing studs, power supplies, amplifiers, signal conditioners.

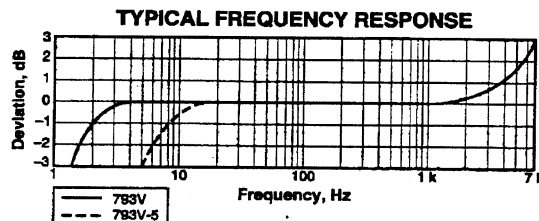
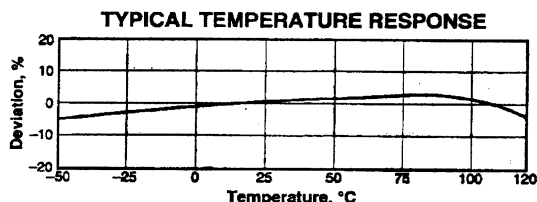


FIGURE 31
793V SERIES PIEZOELECTRIC VELOCITY TRANSDUCERS (Ref. 5)

2.323 Non-contact Eddy Current Displacement Probes

Non-contact eddy current displacement probe systems like that shown in Figures 32 and 33 (sometimes called proximity probes) are used primarily to measure relative shaft vibration, radial and axial shaft position, and the differential expansion between the case and the rotor. Accelerometers and velocity pickups measure absolute vibration. If the overall peak velocity is measured on a bearing cap, this is a measure of its motion. Non-contact pickups are especially effective on large machinery outfitted with oil film plain bearings such as turbine/generators, compressors, large motors, etc. Thus, when ordering such large, high-speed machinery, a plant may wish to include in the specifications non-contact displacement probes that are to be internally mounted close to the shaft in the heavy casing. In this case, it is expedient to measure the vibration directly at the shaft since much of the energy will not transmit through the heavy casings. By comparison, if it was suggested to measure the vibration on the exterior surface of the housing (as it is done typically by PMP personnel on a data collection route) with an accelerometer or velocity pickup magnetically mounted (sensing absolute motion), the vibration energy originating from the shaft can be greatly attenuated by the time it is sensed by the bearing cap transducer. Thus, a non-contacting displacement transducer would be a better selection for these machines, particularly for measurements at operating speed and up through the first five or six harmonics of operating speed.

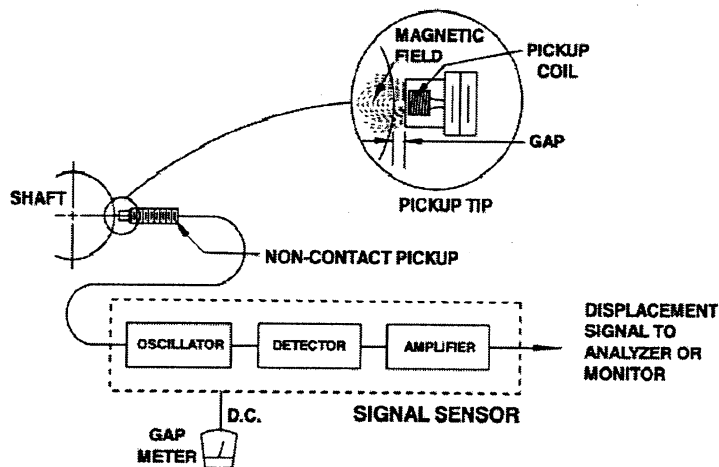


FIGURE 32
SCHEMATIC DIAGRAM OF AN EDDY CURRENT NON-CONTACT DISPLACEMENT PROBE SYSTEM (Ref. 5)

Be aware that the readings from a non-contact eddy current displacement probe (measuring relative shaft vibration) may be quite a bit higher in amplitude than those taken on the casing itself. In the case of a journal bearing machine where the shaft is 3 to 6 feet from the outside, the shaft vibration may be as much as 20 times higher than the reading obtained on the bearing housing. The signal is attenuated as it passes through the oil film, as well as through each of the metal interfaces and thicknesses out to the machine housing. The amount of signal loss depends on the oil-film characteristics, the mass, and the distance from the shaft out to the bearing housing surfaces. This means that potentially serious problems, such as oil whirl or oil whip, may be missed altogether if readings are not taken directly from the shaft itself. Therefore, the best possible vibration spectrum, particularly for events at 1X RPM up through about 6X RPM, is that which can be acquired directly from a non-contact eddy current probe. This non-contact probe will measure the vibration of the shaft relative to the mounting system of the probe itself. Therefore, it is most important to anchor the probe as much as possible to get the truest picture of shaft vibration.

Unlike piezoelectric accelerometers and velocity pickups, a non-contact sensor requires external electronic circuitry to generate a very high frequency AC signal since it does not have an internal component which generates a voltage or an electrical charge in response to the vibration. As shown in Figure 32, this high frequency electrical signal is applied through a coaxial cable to the coil which generates a magnetic field at the tip of the pickup. Then the tip will be placed at a gap of only about 40 to 60 mils from the target surface. The shaft will absorb some of the magnetic energy. As the shaft moves relative to the pickup tip, the probe will provide a peak-to-peak AC signal voltage proportional to the vibration, and a DC signal proportional to the gap. The oscillations in the AC signal caused by vibration of the shaft are detected and can be read by a permanent monitor or an analyzer connected to an output of the non-contact probe to determine the amount of relative shaft vibration while the DC signal can be used to monitor the change in gap. A non-contact probe can be particularly useful if it is targeted on the shaft's end to detect axial shaft movement and possible wear of thrust bearings.

Figure 33 shows a non-contact probe mounted in a bearing housing. The signal from the probe can be taken directly into an analyzer for a measurement. Be very careful! Do not connect the cable to the ICP port on a spectrum analyzer or data collector. The ICP current will cause the vibration "readout" to suddenly jump which can cause these machines to alarm, or even worse, trip it off line.

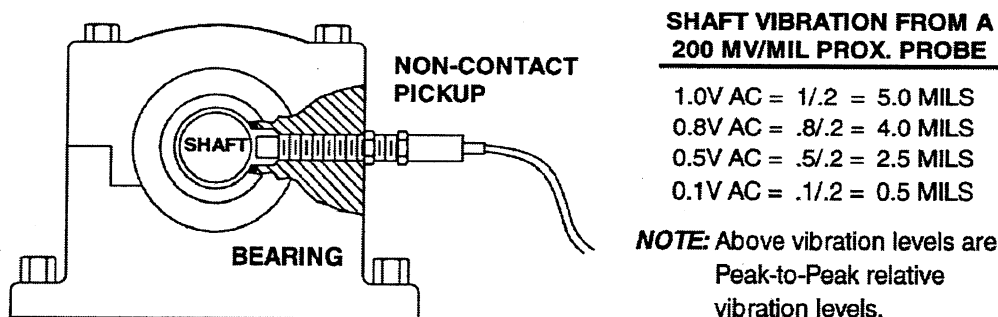


FIGURE 33
NON-CONTACT EDDY CURRENT PROBE MOUNTED IN A
BEARING HOUSING (Ref. 5)

Figure 34 shows a typical response curve for a non-contact probe. This curve represents the change in sensor output voltage as the distance between the tip of the pickup and the target material is altered. Note that the relationship of the output voltage to the gap approximates a straight line. The slope of this line in millivolts per mil (mV/mil) determines the non-contact probe sensitivity (normally non-contact probe sensitivities range from about 100 to 200 mV/mil, peak-to-peak). The system shown in Figure 34 has a linear gap range from about 20 to 100 mils or a total of 80 mils. Therefore, the optimum "setpoint" would be selected at the midpoint (60 mils in this case) of the specified linear gap range. Note that most larger plain bearing machinery (turbines, boiler feed pumps, etc.) are outfitted with "X-Y" probes (with each probe normally located 45° off of vertical). This will be utilized to capture Lissajous Orbits which track actual shaft motion (to be covered in later chapters).

Of great importance when installing non-contact probes is to provide a very smooth, uniform surface finish for the target area. The American Petroleum Institute Standards (API 670) call for a surface finish of 16 - 32 microinches RMS (or 0.40 - 0.80 micrometers RMS). The target area must be free of any surface imperfections such as scratches or high spots. In fact, a common problem with non-contact probes is that, when a scratch is present, the probe cannot discern between true vibration and the scratch depth, thereby causing an error in the readout.

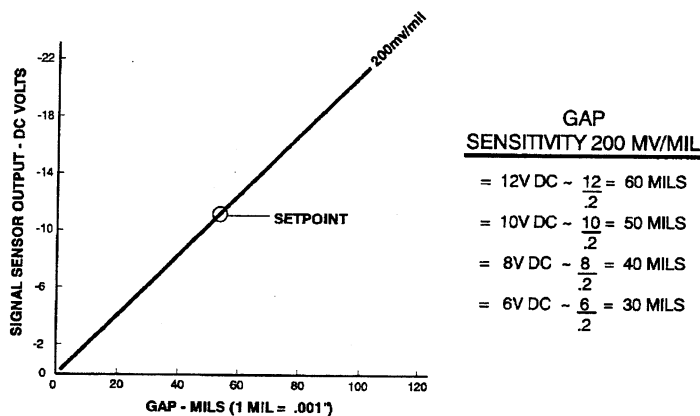


FIGURE 34
TYPICAL OUTPUT FOR AN EDDY CURRENT NON-CONTACT DISPLACEMENT PROBE SYSTEM (Ref. 5)

Another problem common to non-contact probe systems is either mechanical or electrical runout. Mechanical runout is an error which is caused by out-of-roundness and surface imperfections (scratches, chain marks, dents, rust or other conductive buildup on the shaft, flat spots and engravings). It is most important to ensure that there is less mechanical runout in the shaft than the minimum allowable vibration displacement which is to be measured.

Readings should be taken at 30° - 45° rotational intervals around the shaft, plotting runout versus angular position to ensure this, and also to provide a baseline for comparison with electrical runout data. Figure 35 shows a waveform captured from a non-contact probe which was targeted on a shaft having a scratch in the target area. Note the pronounced pulses in the time domain spaced at once/revolution each time the scratch passed the probe.

Electrical runout is a signal error which occurs in eddy current displacement measurements when the shaft surface conductivity varies. That is, the target's surface itself has nonuniform electrical conductivity/resistivity/permeability properties. Another source of electrical runout is due to the presence of a localized magnetic field(s) on the shaft surface. This error will repeat exactly with each shaft revolution. Such shaft magnetism may result from magnetic-particle testing; or by operating in a strong magnetic field produced by motors, generators or alternators; or can even be induced from welding operations nearby. In these instances,

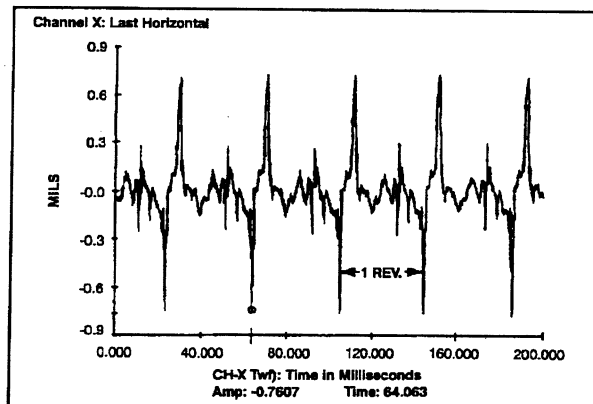


FIGURE 35
TIME WAVEFORM TAKEN FROM A NON-CONTACT PROBE TARGETED ON A SCRATCHED SHAFT (Ref. 10)

it is impossible for the displacement probe to differentiate true vibration from either the mechanical or the electrical runout. Depending on where the electrical and mechanical runouts are located on the shaft, they may either add to or subtract from each other. That is, **the displacement readout will be a vectorial addition of electrical runout, mechanical runout, and the actual vibration.** There are procedures in place for determining the presence of mechanical and electrical runout and for minimizing their effect. In the case of electrical runout, it often is necessary to correct this with a degaussing instrument.

Remember that it is very important to calibrate the non-contact probe for the specific shaft material to which it will be targeted. Figure 36 shows typical response curves for different target materials in comparison to the gap that is set between the probe and the target. This chart is used to set up and calibrate the response of the proximity probe. It is important to notice that "roll-off" problems occur as the gap is increased. It is recommended that probes not be set up in these roll-off ranges.

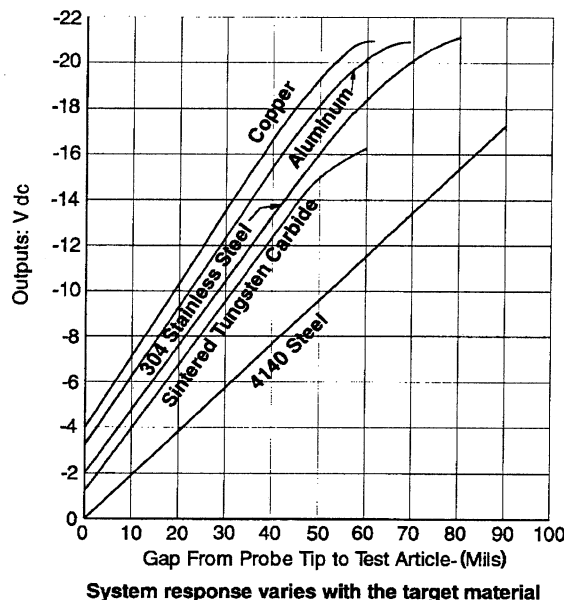


FIGURE 36
VOLTAGE RESPONSE VERSUS GAP FROM PROBE TIP TO SHAFT FOR
VARIOUS TARGET MATERIALS (Ref. 8)

Finally, a remaining problem with a non-contact probe is in the high frequency range. Although specifications for non-contact probe systems will claim a frequency response up to 600,000 CPM, this does not actually occur in the real world. Since the probe is measuring displacement, it will tend to emphasize low frequency sources, but de-emphasize high frequency sources. For example, consider a centrifugal compressor outfitted with non-contact probes on each of its impellers which range in operating speeds from about 20,000 up to approximately 50,000 RPM. Since the displacement amplitudes become very low with increasing frequencies, these displacement measuring systems will not be sufficiently sensitive to pick up problems with gear wear or with failing thrust bearings on these machines (which are commonly outfitted with about 6 pads on each thrust bearing). However, if problems such as unbalance (due to impeller buildup and/or erosion) or misalignment occur, the system will readily respond, whether it is targeting a low-speed or a high-speed impeller. Still, however, such machines should always be outfitted with non-contact probe permanent vibration monitoring systems in order to detect the actual shaft vibration and the onset of problems to which they are in fact sensitive.

2.324 Shaft Contact Displacement Probes

Shaft contact probes (shaft sticks and shaft riders) are probes that directly measure shaft vibration by actually riding on the shaft surface. Although they have been replaced by non-contact eddy current probes in most cases today, they are still used in cases where a proximity probe has not been installed, for balancing or periodic vibration checks. These transducers are limited in frequency response to about 12,000 CPM (200 Hz) and should not be used for shaft surface speeds over approximately 15,000 ft/min (approximately 5000 m/min).

2.3241 Shaft Sticks

A shaft stick should only be used if a non-contacting displacement probe has not been installed in a large journal bearing machine. Figure 37 is a picture of a shaft stick in use. It consists of a hardwood, "fish-tailed" stick outfitted with a stud on its other end to attach a velocity pickup or an accelerometer. The fish-tail shape helps keep the shaft stick in contact with the shaft and the edge of the fish tail is tapered to reduce friction and prevent chatter. The shaft stick is hand-held perpendicular to the shaft.

When using the shaft stick, use special care to follow these precautions:

- Do not mount the stick on shafts traveling faster than approximately 15,000 ft/min (5000 m/min) for safety reasons.
- Try to use the same pressure from one survey to the next when taking PMP type measurements as this can affect your month to month vibration amplitudes.
- Do not rely on the shaft stick to monitor "high frequency" defects such as bearings, blade passes, gear mesh frequencies, and certain electrical problems.

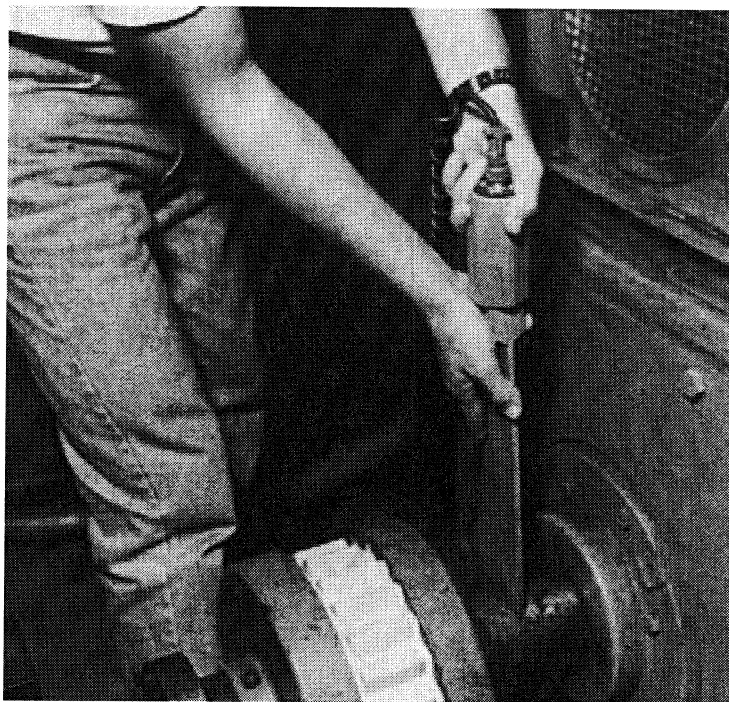


FIGURE 37
SHAFT STICK (Ref. 1)

2.3242 Shaft Riders

A shaft rider (pictured in Figure 38) is mounted permanently in the housing and extends into the casing to measure shaft displacement by using a spring mounted tip that rides on the shaft's surface. One of two methods is usually used. In the first, an accelerometer or velocity transducer is mounted on the end of the shaft rider (quite similar to using an accelerometer or velocity transducer with a probe). In the second, a spring-loaded rod is driven by the vibration and moves a mass inside coiled wires surrounded by magnets. An electrical signal which is used to measure the vibration is thereby generated.

Once again, the shaft rider is an older technology which has gradually been replaced by the non-contact eddy current probe, but is still in place on many of today's turbine/generators. Typically, the shaft rider is employed to continually monitor the vibration of the shaft in order to determine if the vibration is at a level which would signal that a machine should be shut down or that an alarm should be sounded. Like the shaft stick, its usable frequency range is up to about 12,000 CPM. Furthermore, it should be used only on machines that do not have shaft speeds which exceed 3600 RPM and a shaft surface speed below approximately 20,000 ft/min. Thus, the shaft rider is useful only for monitoring unbalance, misalignment, and other problems occurring at 1X thru 3X RPM. It will not be able to detect rolling element bearing wear or failures, gear mesh problems, and other high frequency problems.

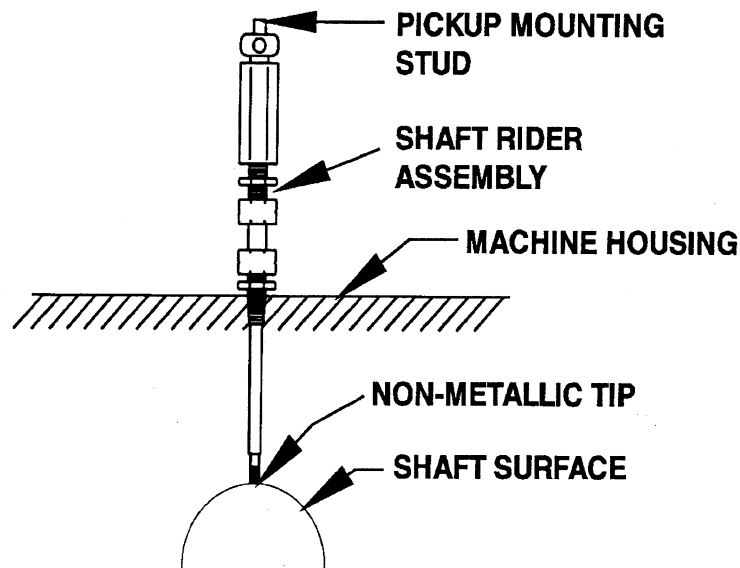


FIGURE 38
SHAFT RIDER (Ref. 5)

2.33 SELECTION CRITERIA FOR TRANSDUCERS

Below are the descriptions of the items in Table III, plus other items which should be considered when selecting a transducer:

- 1) **SENSITIVITY RANGE** - Sensitivity is the capability of the transducer to determine the amplitude of vibration (displacement, velocity, or acceleration) from the amplitude of the voltage signal. For example, an accelerometer may have a sensitivity of 100 mV/g. This means that if this 100 mV/g accelerometer saw 10 mV at a frequency, it should convert this voltage to an amplitude of 0.1 g ($10/100 = 0.1$).
- 2) **FREQUENCY RANGE** - Frequency range is a measuring capability of the transducer from a low limit to a high limit of frequency. Each transducer has its own frequency range (which must be known by the user). Typically, the transducer's frequency response is given at various amplitude tolerances, such as $\pm 5\%$, $\pm 10\%$, and/or $\pm 3\text{dB}$.
- 3) **NATURAL FREQUENCY RANGE** - The natural frequency of the transducer is the factor that limits the upper end of the frequency range of the transducer. Therefore, it is very important to choose an accelerometer in which the highest frequency of interest is well below its natural frequency. Normally, the effective accurate range of its frequency will be about 20% - 33% of its stud-mounted natural frequency. Thus, for an accelerometer with a natural frequency of 30,000 Hz (1,800,000 CPM), the effective range for the upper frequency will be limited to about 6000 - 10,000 Hz (360,000 - 600,000 CPM).
- 4) **TRANSDUCER MASS** - The weight of the transducer must be only a fraction of the weight (usually below 10%) of the housing to which it will be attached. If the transducer is heavy enough to affect the mass, it can change the natural frequency of the system, and therefore, the true frequency response.
- 5) **USABLE TEMPERATURE RANGE** - This is the minimum and maximum temperature that a transducer can withstand without significantly affecting its response capabilities. This is especially important when selecting transducers to be mounted permanently on machinery that is subjected to very high or very low temperatures. Also, please see the important comments in Section 2.321 pertaining to how differently shear mode accelerometers respond to temperature *differences* than do compression mode accelerometers, particularly when taking low frequency measurements (see these accelerometer types in Figures 17 thru 19).
- 6) **MEASUREMENT DIRECTION** - Most transducers measure only in the mounting direction (with only a small percentage reaction to vibration in directions perpendicular to the mounting - typically 3% to 6%). However, triaxial transducers measure vibration in all three directions simultaneously.
- 7) **TRANSDUCER SIZE** - The measurement location on a machine may require a transducer to be of a certain size or cross section to physically fit on the measuring surface. For instance, on a small surface, a small transducer must be used. However, due to its small size, the transducer may have a low voltage sensitivity restricting its capability to pick up low frequency data.

-
- 8) **TRANSDUCER POWER SUPPLY** - There are 3 common methods to power a transducer:
- i) ICP¹⁵ (Integrated Circuit Piezoelectric) Power Supply. The vibration instrument sends the power through the cable to the transducer.
 - ii) Independent Power Supplies. A battery (DC) or AC-powered unit is used to send power to the transducer.
 - iii) Charge Amplifier. A battery or AC powered unit is placed between the transducer and the vibration instrument to amplify the signal voltage.
- 9) **CABLING** - ICP powered transducers can generally use cables up to about 1000 feet in length. For transducers which are not ICP powered, a charge amplifier may be required at approximately 50 foot intervals of cable length (or less). Discuss the cabling requirements for the transducer with the transducer vendor.
- 10) **MOUNTING SENSITIVITY** - As described in **Sections 2.34 and 2.341**, there are many ways to mount transducers (hand-held probes, magnetic connectors, permanent stud mounts, adhesive mounts, etc.). Each has a significant effect on the ability of the transducer to measure the vibration accurately, as well as to reproduce the same spectra in subsequent measurements. **This one fact is often critical to obtaining accurate, repeatable data (and, therefore, to the success or failure of the entire PMP program).**
- 11) **MAGNETIC INTERFERENCE** - Except for piezoelectric accelerometers, magnetic interference affects the performance of transducers. Since eddy current probes rely on a magnetic field to determine shaft gap, the influence from another magnetic field will produce erroneous readings. Likewise, moving coil seismic velocity pickups can be effected by magnetic fields. Even cable movement generates a magnetic field that can distort the signal in the cable. The magnetic field around large generators and motors can affect seismic transducers, cables, and even the analyzer itself.
- 12) **SEALING METHOD** - *If they are not certified as being hermetically sealed, all accelerometers will "micro-breathe", allowing moisture to affect their lower frequency (below 3000 CPM or 50 Hz) performance dramatically.* If the sealing method is not listed in the specifications, contact the accelerometer vendor and verify.

2.34 MOUNTING OF TRANSDUCERS (ACCELEROMETERS)

There are 5 typical transducer mounting methods, as listed in Table IV. Each mounting method will provide a typical frequency response range when transducers capable of operating within these frequency response ranges are used.

TABLE IV
TRANSDUCER MOUNT USABLE FREQUENCY RANGE FOR THE
WILCOXON 726T ACCELEROMETER (Ref. 9)

ACCELEROMETER MOUNTING	MAXIMUM ACCEPTABLE FREQUENCY (CPM)	MOUNTING NATURAL FREQUENCY (CPM)
1) Stud Mount	975,000	1,900,000
2) Adhesive Mount with Hottinger Baldwin Messtechnik X60	540,000	None Observed
3) Stud Mount on Rare Earth Magnet	450,000	724,500
4) Mounted on Quick-Connect Stud Mount	360,000	609,000
5) Hand-held Mount Using a 2" Probe	48,000	88,500

Table IV is a summary of an article written by Computational Systems, Inc. (CSI)⁹. Many other factors come into play when the effectiveness of a transducer mounting type is considered. Refer to the referenced article for an in-depth look at transducer mounting effectiveness. Figure 39 shows illustrations of these mounting methods (Quick-Lock not shown).

2.341 Transducer Mounting Applications

Each particular transducer mounting method has certain applications which will be discussed in detail. Following below is a description of each mount, along with some important comments which should be made (remember that an incorrect transducer mount, such as use of a hand-held probe can corrupt the data you are trying to acquire, and can even miss very important high frequency data pertaining to gears, rolling element bearings and electrical problems):

STUD MOUNT - Stud mounting is used for permanently mounted transducer applications. Sometimes, an adhesive will be used in combination with stud mounting to prevent the transducer from working its way off of the stud mount. While stud mounting is not practical for collecting PMP route data, if very high frequency measurements (>3,000,000 CPM or 50,000 Hz) are required, it will have to be used at those data collection points. Also, stud mounting gives extremely repeatable data from measurement to measurement from one survey to the next and provides the highest frequency response range.

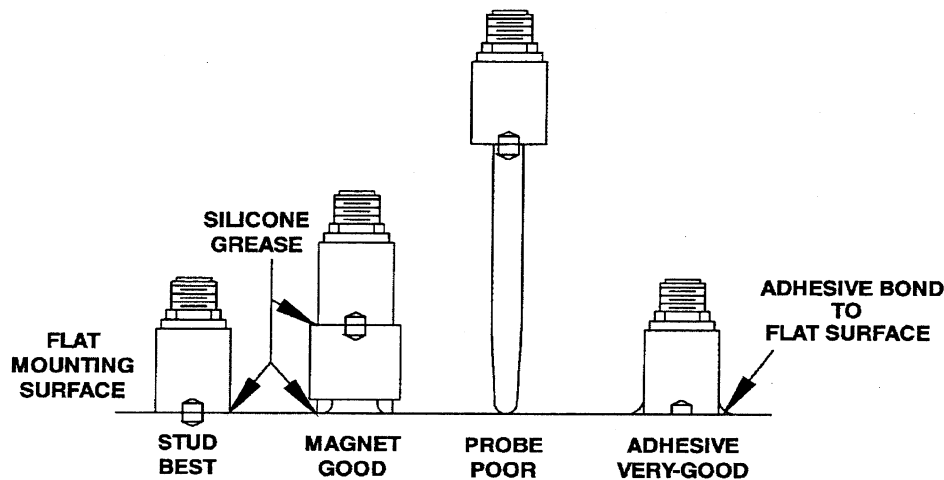


FIGURE 39
MOUNTING METHODS ILLUSTRATED (Ref. 1)

ADHESIVE MOUNT - Adhesive mounts are very useful in diagnostics where reliable high frequency data is required since a stud mount is rarely found at an exact point where a measurement is needed. Occasionally, adhesive mounts are used on a PMP route to collect high frequency data (>2,400,000 CPM or 40,000 Hz). Therefore, great care should be taken when using an adhesive to permanently mount a transducer because these mounts may work loose over long periods of time. Adhesive mounts provide a very good frequency response range if the proper adhesive type is used and if only a thin layer of adhesive is applied. Also, adhesive mounting gives very repeatable data over a series of measurement surveys.

MAGNETIC MOUNT - Magnetic mounts are the most common used method in PMP programs today as well as when taking diagnostic data. The upper frequency response range is generally from about 120,000 to 450,000 CPM (depending on the magnet type, its pulling force, and especially how well the magnet mounts on the surface). Because some machines such as centrifugal compressors have gear mesh frequencies and harmonics that range from 900,000 to 4,000,000 CPM (15,000 - 65,000 Hz), data on these machines must be taken with an adhesive or stud mount instead. However, a magnetic mount tends to give reliable repeatability over measurement surveys which is adequate for PMP purposes when used within the frequency ranges applying to that particular magnet (it is most important to verify this with the vendor supplying the transducer and magnet). Also, frequency response of a magnet mounted accelerometer can be greatly enhanced at frequencies exceeding approximately 90,000 CPM (1500 Hz) if one takes 10 to 15 seconds to insert silicone grease between the magnet and the accelerometer. This was clearly illustrated in the *Sound and Vibration* magazine article authored by Dr. Ken Piety *et al* listed in Reference 9.

QUICK-CONNECT MOUNT - Quick-Connect mounts are also ideal for collecting PMP route data since they provide easy mounting and dismounting. They have a relatively large frequency range which is able to detect most common machinery problems which occur in higher frequencies such as bearing defects. However, a quick-connect mount should not be used for detecting high frequencies above approximately 420,000 CPM such as those found in high speed compressor gear mesh frequencies and harmonics (see Ref. 9). Nevertheless, repeatability between measurement surveys is consistent enough for PMP purposes for frequencies below this limit.

HAND-HELD PROBE - This is the most undesirable mounting method. The usable frequency range is only up to a maximum of 60,000 CPM, regardless of the length, diameter or material of the probe! (as per Ref. 9). Depending on the length of probe used, the maximum accurate frequency measurable may only be 30,000 CPM. With these constraints, the only machinery problems detectable and repeatable will normally be those which occur at 1X, 2X, and 3X RPM for machinery running under 3600 RPM or less. Most importantly, bearing defect frequencies and their harmonics, and bearing component natural frequencies often lie above 30,000 CPM. They can be missed entirely if a hand-held probe is used. Furthermore, as demonstrated in the research covered in Reference 9, the repeatability of measurement amplitudes and maximum frequency ranges from survey to survey is not consistent and will vary depending on the probe position or amount of hand pressure applied as well as on which person is taking the data. This impairs the analyst's ability to accurately trend problems and may cause him to miss some problems that occur at the top (or beyond) of the probe's measurable frequency range. Hand holding a transducer or probe is useful in hard-to-reach places such as where a screen prevents the use of a magnet or other mount type. Also, it is useful if safety is a concern and the analyst cannot safely reach into moving parts of the machine. However, be aware that using this method will miss very important information that is crucial to maintaining a successful PMP program. In the instances described above, it is wiser to attach a permanently mounted transducer in unsafe or hard to reach locations to obtain useful diagnostic data. Furthermore, this example shows that probes actually can add "phantom" data in the bearing defect range which makes the bearing appear to be in its last stage of failure due to the natural frequency of the probe being excited even though there may be no bearing wear at all. And, what might be worse is that Reference 9 proved even the short 1/2 inch probes will almost always totally miss data at frequencies exceeding 120,000 CPM (2000 Hz) which again is where many significant gear, rolling element bearing and electrical problems are so often detected (sometimes with little or no evidence of such problems below this frequency).

2.4 UNDERSTANDING VIBRATION PHASE AND ITS APPLICATIONS

2.41 INTRODUCTION

Phase is the timing relationship vibration has with respect to another vibrating part or fixed reference point. It can also be thought of as the vibration motion at one location relative to the vibrating motion at another location (for example, phase at the outboard bearing in the horizontal direction relative to that in the vertical).

Phase is easiest to visualize if one is familiar with using a timing light to set the timing of an automobile engine. Vibration phase is very similar except that the vibration (not the spark) is the trigger. *If an analyst clearly understands phase, he can use this powerful analytical tool to differentiate between the many problems which can cause high vibration at 1X RPM, 2X RPM, and 3X RPM with conviction.* However, the vibration frequencies have to be the same or exact multiples to properly compare phase differences. By taking phase measurements on each bearing housing in each direction (horizontal, vertical and axial), he can determine whether the problem is unbalance, misalignment, soft foot, bent shaft, eccentric rotor, loose hold-down bolts, resonance, cocked bearing or several other potential problems (all of which can generate vibration spectra which appear to be *identical*). In a later chapter section, it will be shown that a trigger will be required to serve as a reference when using a single-channel analyzer to acquire phase measurements.

When should phase measurements be taken? This section, along with the chapter describing each of the items in the "Illustrated Vibration Diagnostic Chart" will show that *phase measurements should be taken when high vibration is found at 1X RPM, 2X RPM and/or 3X RPM.* In other words, phase measurements are not intended to be taken at each point on each machine during routine data collection.

2.42 HOW TO MAKE PHASE MEASUREMENTS

Figures 40 and 41 illustrate two typical strobe light methods for taking phase measurements (strobe light for use with data collectors are now available). Another method, shown in Figure 42, utilizes a stationary photocell or laser-tach targeted at a piece of reflective tape mounted to the rotating part.

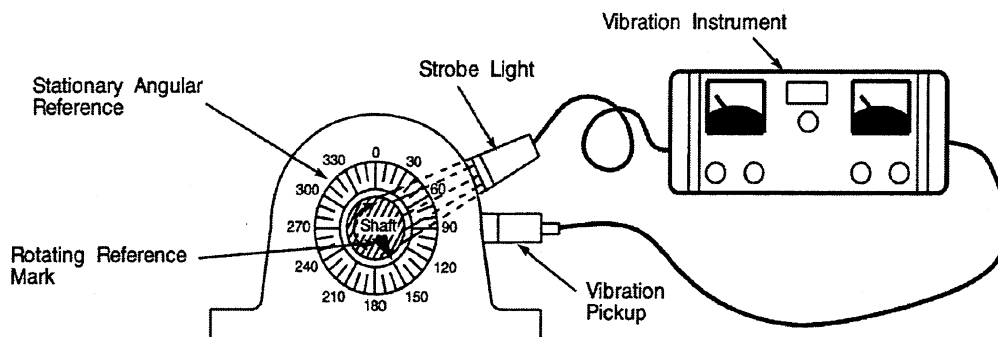


FIGURE 40
PHASE REFERENCE WITH A ROTATING REFERENCE MARK AND A
STATIONARY ANGULAR REFERENCE

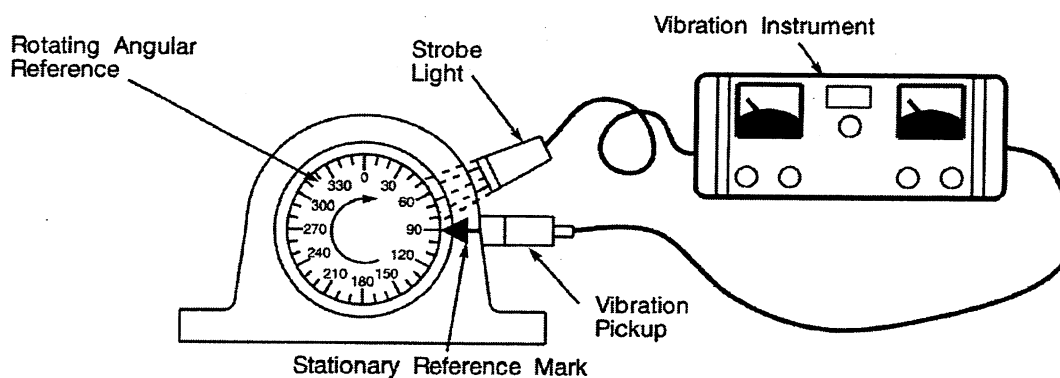


FIGURE 41
PHASE MEASUREMENTS WITH A STATIONARY REFERENCE MARK AND
ROTATING ANGULAR REFERENCE

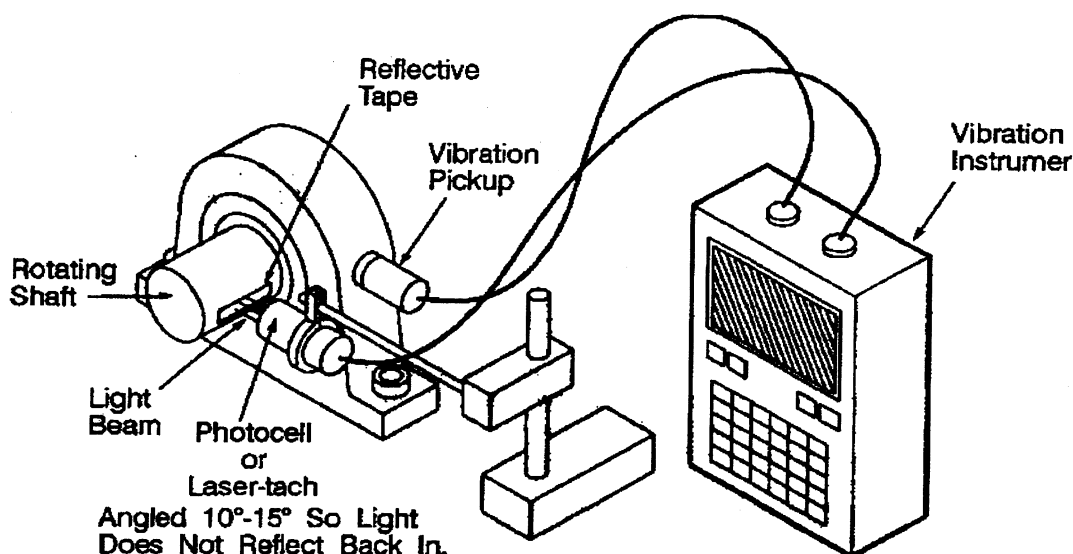


FIGURE 42
PHOTOCELL METHOD OF ACQUIRING PHASE MEASUREMENTS

To properly collect phase measurements, it is important that the following be done first. After the strobe light is "tuned" to the frequency of interest, the vibration amplitude and phase must be recorded. If using a swept-filter instrument like that shown in Figure 40, the strobe light can be oriented in any way to best see the reference mark, but the transducer must be firmly mounted in place at each measurement point. *In contrast, if a photocell is used, both the photocell and the transducer must be locked down at each measurement point.* Only the transducer gets moved to the next location.

The photocell method is more accurate than the swept-filter strobe method since the instrument measures the phase angle within very accurate tolerances. Since the strobe method is hand-held, it includes human error.

2.43 USING PHASE ANALYSIS IN VIBRATION DIAGNOSTICS

To determine whether the whole bearing face is moving back and forth axially (as in Figure 43), or if it is vibrating with a twisting motion (as shown in Figure 44), phase measurements are taken from four points on the bearing housing in the axial direction.

2.431 Evaluating Axial Motion of a Bearing Housing to Reveal a Possible Cocked Bearing or a Bent Shaft

Using the conventions shown in Figure 43, the transducer should be mounted at locations 1, 2, 3, and 4. In this case, the phase analysis indicates that the bearing is moving axially in a planar motion.

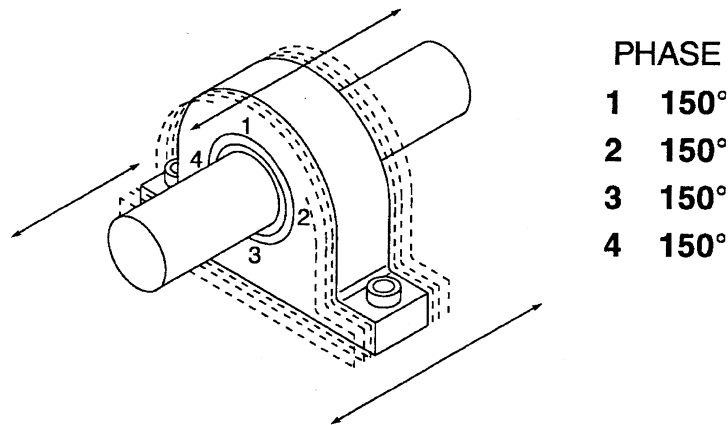


FIGURE 43
AXIAL PHASE SHOWING PLANAR MOTION

However, if the same four measurements produce a 90° phase difference at each location, as shown in Figure 44, a bent shaft or a cocked bearing would be suspected (particularly if confirmed with axial phase measurements on the other bearing of the same rotor as in Figure 45. In this case, the 180° phase change across points 1 and 3 indicates an up-and-down twisting motion while the 180° difference between points 2 and 4 reveals a side-to-side twist.

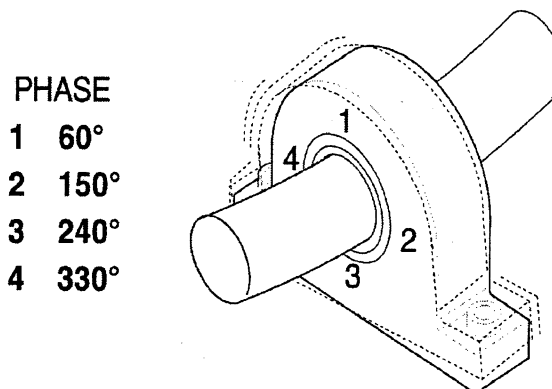


FIGURE 44
AXIAL PHASE SHOWING TWISTING MOTION DUE TO BENT SHAFT
OR COCKED BEARING

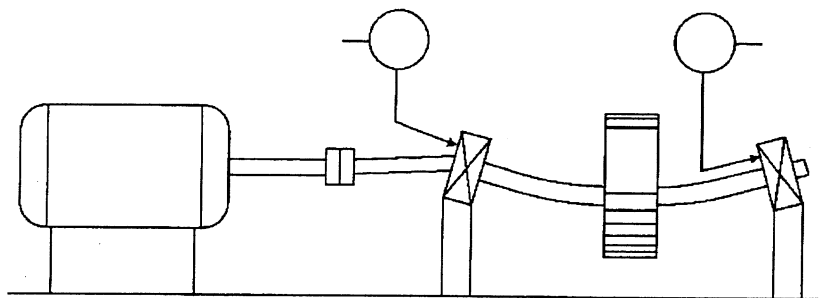


FIGURE 45
DETECTION OF BENT SHAFT PROBLEMS

Bent shaft problems cause high axial vibration with axial phase differences tending toward 180° on the same machine component (that is, between outboard and inboard bearings of a motor or between the driven rotor bearings themselves). Dominant vibration normally occurs at 1X if bent near shaft center, but at 2X if bent near the coupling. (Be careful to account for transducer orientation for each axial measurement if you reverse transducer direction.)

2.432 Phase Behavior Due to Unbalance

To check for an unbalance condition, use the radial phase measurements, as illustrated in Figure 46. If an unbalance condition is the only problem a machine has, the phase increments between each of the radial measurement locations on each bearing will be 90° apart. If the transducer is moved 90° at a time, significant departure from 90° phase difference means that the problem is something other than unbalance. An even better indicator of unbalance is to compare the difference in phase in the horizontal direction on the outboard and inboard bearings. Then, find the vertical phase difference between measurements on the outboard and inboard bearings. If the unbalance is significant, the horizontal phase difference should equal the vertical phase difference between the outboard and inboard bearings within $\pm 30^\circ$. Also, the 1X RPM peak will be high. This shows that the resultant motion of the rotor is the same in both the horizontal and vertical directions. If the resultant motion is not the same, the dominant problem is something other than unbalance.

2.433 Phase Behavior Due to Looseness/Weakness

If the amplitude and/or phase changes appreciably between the mated components, as shown in Figure 47, looseness/weakness should be suspected. Note the significant amplitude and phase change between the measurements on the baseplate and the supporting concrete base. This problem may be caused by inadequate grouting between these two surfaces. When phase changes across any two interfaces (i.e., between the base plate and concrete base) exceed approximately 90° , such a problem is usually indicated. Also, look for great differences in amplitudes between each interface (although this can occur if there is considerably more mass and, therefore, less vibration response at one of the interfaces).

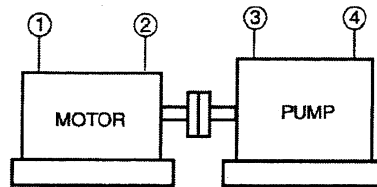


TABLE A.
PHASE AT 1X RPM FOR A MACHINE
WITH DOMINANT FORCE UNBALANCE

DIR.	1	2	3	4
A	60°	*70°	60°	*80°
H	30°	25°	30°	40°
V	120°	110°	120°	135°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

TABLE B.
PHASE AT 1X RPM FOR A MACHINE
WITH DOMINANT COUPLE UNBALANCE

DIR.	1	2	3	4
A	60°	*70°	60°	*80°
H	30°	210°	200°	220°
V	120°	295°	280°	300°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

TABLE C.
PHASE AT 1X RPM FOR A MACHINE
WITH SIGNIFICANT DYNAMIC UNBALANCE

DIR.	1	2	3	4
A	60°	*70°	60°	*80°
H	30°	90°	80°	70°
V	120°	180°	170°	165°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

FIGURE 46
TYPICAL PHASE MEASUREMENTS WHICH WOULD INDICATE EITHER
STATIC, COUPLE OR DYNAMIC UNBALANCE

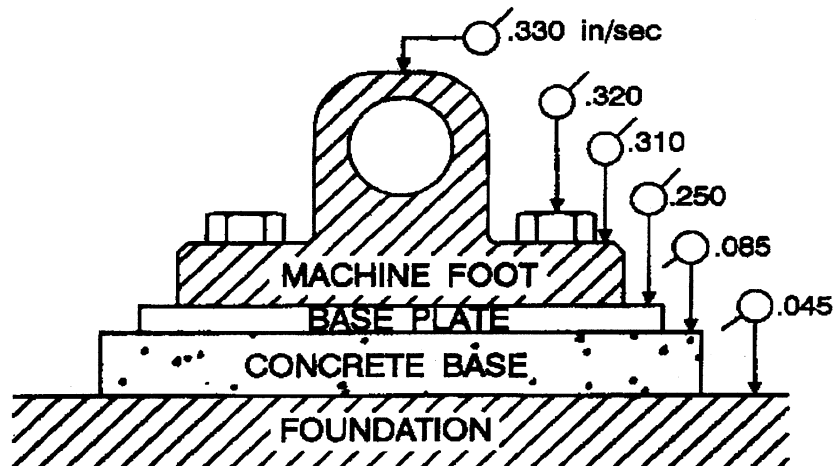


FIGURE 47
PHASE MEASUREMENTS FOR LOOSENESS

2.434 Phase Behavior Due to Misalignment

Figure 48 illustrates the two basic types of shaft misalignment.

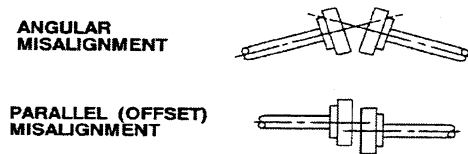
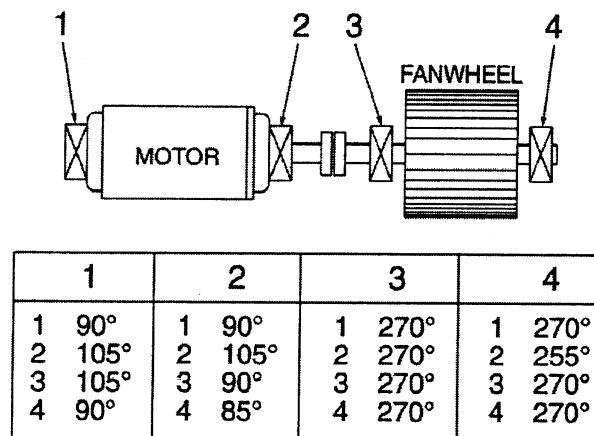


FIGURE 48
DIAGRAM OF ANGULAR AND PARALLEL (OFFSET) SHAFT MISALIGNMENT

Most misalignment is rarely pure angular, or pure parallel, but usually a combination of both instead. Shaft misalignment is characterized by phase differences approaching 180° across the coupling. Phase differences are not exactly 180° out-of-phase or 0° in-phase, but usually within 30° of each other. The closer the phase difference is to 180° , the higher is the probability of misalignment. Radial phase readings are normally more sensitive to parallel misalignment and axial phase readings to angular misalignment.

Amplitude and phase measurements should be taken both axially and radially on both bearing housings on both sides of the coupling. Then, the phase readings should be compared - first, between the bearing housings on each side, then between the housings on each side of the coupling itself. A phase difference of 180° across the coupling strongly indicates shaft misalignment.

To check for angular misalignment in particular, readings should be taken in the 4 quadrant locations like the conventions shown in Figures 43 and 44 (or in as many of the 4 points as accessible). An example of this is given in Figure 49. Consider the data which is provided. Motor Bearings 1 and 2 are moving in unison with one another while Bearings 3 and 4 are also moving together (this indicates there is no internal misalignment, bent shaft, or cocked bearings in either the motor nor the fan). However, notice the 180° phase change across the coupling. This indicates that the motor and fan shafts are misaligned between Bearings 2 and 3, and show that, at any instant of time, the motor shaft is always moving in a direction opposite to that of the fan shaft in the axial direction.



Note: Phase measurements have been corrected for transducer direction.

FIGURE 49
AXIAL PHASE COMPARISONS FOR ANGULAR MISALIGNMENT

Since the transducer often must be turned 180° when making some of the axial measurements, remember to add or subtract 180° from the phase reading each time it is oriented opposite the reference direction (or viewing) that was chosen, so that the phase readings are forced to orient in the same reference direction.

2.435 Using Phase Analysis to Find the Operating Deflection Shape of a Machine and Its Base

Acquisition of the operating deflection shape can go far in helping an analyst diagnose both the cause and severity of problems on a machine or support structure. As the name implies, this technique reveals how the member is vibrating at a certain frequency. Importantly, operating deflection shape analysis is not equivalent to a modal analysis since it does not require acquisition of the frequency response functions (FRF's) of a machine member. And, it can be performed using only a once per revolution trigger and a single channel instrument (even a swept-filter analyzer and strobe light). While it often may reveal resonant operation, it just as often can detect misalignment, soft feet, etc.

The operating deflection of a machine and/or its support structure can be determined by dividing the structure into 10 or 12 equally-spaced measurement locations and recording the phase and vibration values at each location. The objective is to determine the location of the maximum and minimum flexure. With this information, coupled with the forcing frequency (1X RPM, 2X RPM, or at 6X RPM in the case of a 6-vaned pump impeller), a solution can be derived to reduce the excessive motion if resonance is found by adding braces or mass, by changing the forcing frequency (or by replacing anchor bolts or correcting "sprung feet" if excessive vibration is found in a support foot or two).

Figure 50 shows some equally spaced phase measurement locations on a machine support frame which had an apparent flexing problem at 1X RPM in the vertical direction of the machine mounted on it. If only the vibration amplitude was used without the phase information, a plot of the vibration amplitude on a drawing of the frame structure may appear like the one shown in Figure 51.

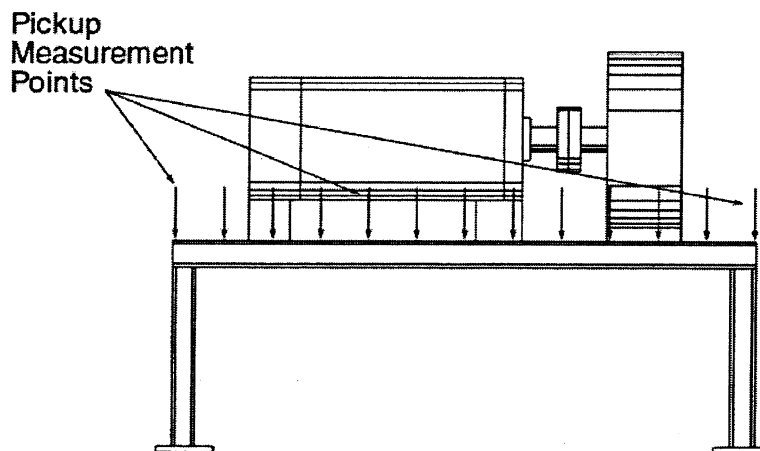


FIGURE 50
PHASE MEASUREMENT LOCATIONS ON FLEXING MACHINE BASE USED
TO DETERMINE ITS OPERATING DEFLECTION SHAPE

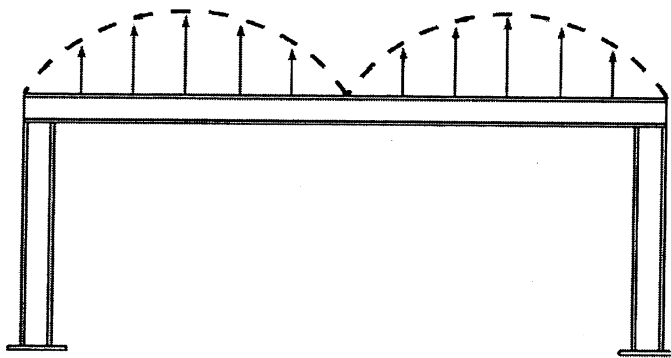


FIGURE 51
PLOTTING ONLY AMPLITUDE WITHOUT PHASE WILL CREATE AN
INCORRECT OPERATING DEFLECTION SHAPE

However, by including the phase information, the true operating deflection shape is shown in Figure 52. The phase data verifies that the left and right halves of the structure were actually moving in opposite directions. Also, note the significant phase change of 165° ($270^\circ - 105^\circ$) measured just to either side of mid-span. In this particular case, this was a strong indication the support frame was being excited at its second bending mode. This was subsequently verified by a series of natural frequency tests.

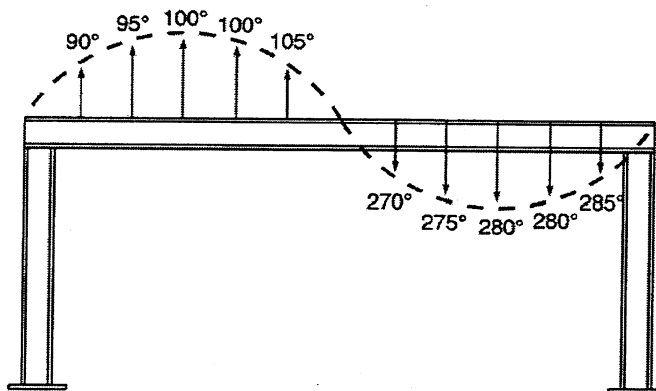


FIGURE 52
ADDITION OF PHASE CREATES AN ACCURATE OPERATING
DEFLECTION SHAPE DRAWING

This type of structural analysis is very important because it shows that stiffening the center of the span, as shown in Figure 53, would have very little effect since it is at a node where the vibration is minimum.

A more effective location for such bracing would be at an antinodal point where the vibration was the greatest, as shown in Figure 54. This bracing was added at the location shown in Figure 54. As a result, the vertical second bending mode increased in frequency by 16%. As a result, vibration levels dropped dramatically from approximately .82 in/sec to only .12 in/sec peak (-85%).

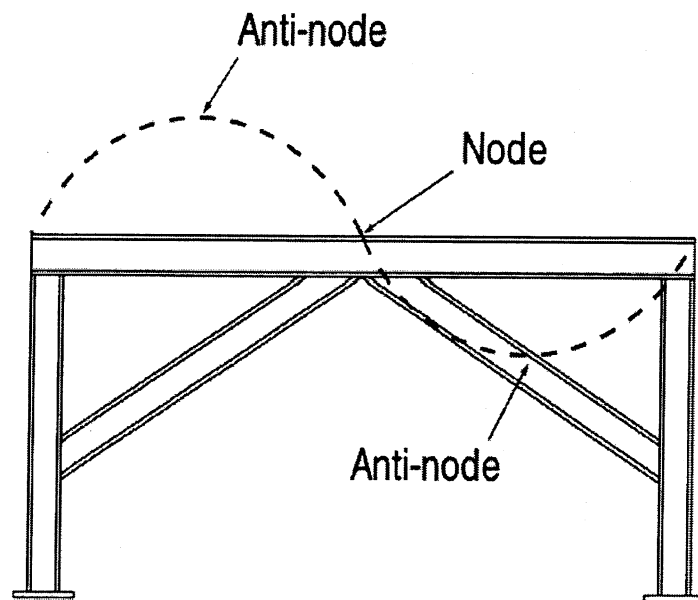


FIGURE 53
INEFFECTIVE APPLICATION OF STIFFENING AT A NODE

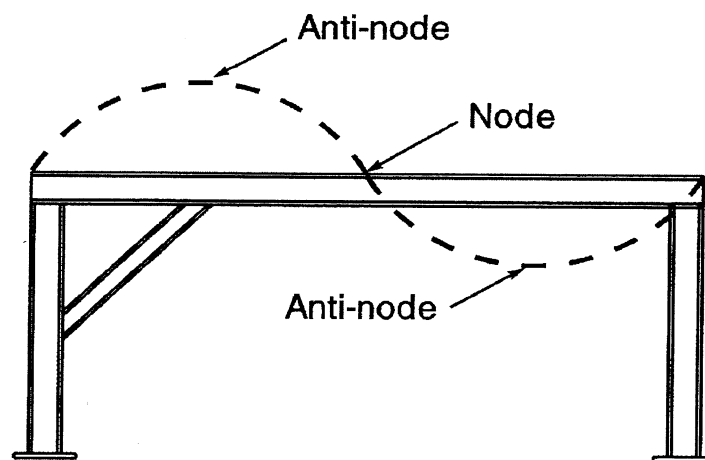


FIGURE 54
STIFFENING AT THE ANTI-NODE AS DETERMINED BY THE MODE SHAPE

Operating deflection shape software is now available which will automate the measurement process and allow a computer to animate the operating shape on a monitor. This makes this technique much more powerful, particularly when attempting to show the results to a person who is not experienced in vibration analysis.

REFERENCES

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11. "Spike Energy Spectrum" - developed by IRD Mechanalysis, Inc., based in Columbus, OH.
12. "Amplitude Demodulated Spectrum" - developed by Computational Systems, Inc. (CSI), based in Knoxville, TN.
13. "Acceleration Enveloped Spectrum" - developed by SKF Condition Monitoring with headquarters in San Diego, CA.
14. "SEE Spectrum" - developed by SKF Condition Monitoring with headquarters in San Diego, CA.
15. "ICP" is a trademark established by PCB Piezoelectronics, headquartered in Depew, NY. "ICP" stands for "Integrated Circuit Piezoelectric".

APPENDIX
**SPECIFICATIONS FOR VARIOUS TRANSDUCERS FROM A
VARIETY OF MANUFACTURERS**

Model 970

Vibration Pickup

SPEC SHEET: 41-06-04
EFFECTIVE: May 1992

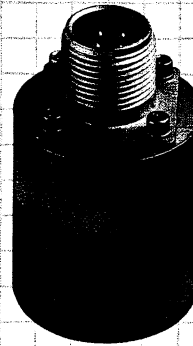
DESCRIPTION

The IRD Model 970 is a rugged low impedance accelerometer type vibration pickup for use with portable instruments on all types of rotating machinery.

The Model 970 has built-in electronics to provide low impedance output and eliminate the need for special low noise cables. The Model 970 is sealed against dust and moisture and operates up to a temperature of 250° F (121°C).

A single 1/4-28 mounting stud is used for attaching the pickup to a flat machine surface. Insulated Mounting Adapters are available for those cases where ground potential problems may exist between the pickup mounting surface and instrument.

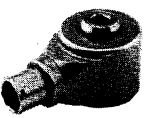
The Model 970 connects directly to IRD Model 810, 840, 880 and other IRD instruments designed to accept the 970. Operating power is obtained from the instrument.



Internally Amplified Vibration Sensors

INDUSTRIAL, GENERAL PURPOSE ACCELEROMETERS

These rugged, general purpose sensors are ideal for machinery monitoring and predictive maintenance applications. They are used on machinery monitoring systems worldwide.

	Sensitivity, mV/g	Acceleration Range, g	Frequency Response, Hz ±5%, ±10%	Temperature Range, °C	Weight, grams	Case Material	Mounting
756 	100	.0007 - 50	3 - 1,000 2 - 5,000	-50 to 90	180	stainless steel	three #8 x 1/8" screws
762 	10	.0015 - 500	3 - 10,000 2 - 12,000	-50 to 120	80	stainless steel	1/4-28 tapped hole
766 	100	.00055 - 50	3 - 10,000 2 - 12,000	-50 to 120	80	stainless steel	1/4-28 tapped hole
768* 	500	.00002 - 10	3 - 4,000 2 - 6,000	-50 to 120	90	stainless steel	1/4-28 tapped hole
776 	100	.0002 - 40	3 - 2,500 2 - 5,000	0 to 85	32	stainless steel	1/4-28 tapped hole
793 	100	.0007 - 50	3 - 5,000 2 - 7,000	-50 to 120	110	stainless steel	1/4-28 tapped hole
793L* 	500	.00001 - 10	0.6 - 700 0.4 - 1,000	-50 to 120	135	stainless steel	1/4-28 tapped hole
793LS 	500	.00001 - 10	0.6 - 700 0.4 - 1,000	-50 to 120	135	stainless steel	1/4-28 tapped hole
793S 	100	.0007 - 50	3 - 5,000 2 - 7,000	-50 to 120	110	stainless steel	1/4-28 tapped hole
796 	100	.0008 - 50	3 - 5,000 2 - 7,000	-50 to 120	100	stainless steel	1/4-28 captive screw

* Ultra low noise sensor.

Internally Amplified Vibration Sensors

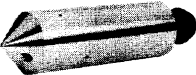
LIGHTWEIGHT, COMPACT SENSORS

Lightweight, compact sensors are used in many laboratory research and light industrial applications. Their low mass makes them effective in applications where transducer mass must be minimized and high signal to noise ratio must be maintained.

	Sensitivity, mV/g	Acceleration Range, g	Frequency Response, Hz ±5%, ±10%	Temperature Range, °C	Weight, grams	Case Material	Mounting
722/722T 	10	.0002 - 500	3 - 10,000 2 - 12,000	-50 to 120	34	stainless steel	10-32 tapped hole
726/726T 	100	.0001 - 50	3 - 10,000 2 - 12,000	-50 to 120	34	stainless steel	10-32 tapped hole
728/728T* 	500	.00003 - 10	3 - 4,000 2 - 6,000	-50 to 120	44	stainless steel	10-32 tapped hole
732A/732T 	10	.0005 - 250	3 - 15,000 1 - 25,000	-50 to 120	15	stainless steel	10-32 tapped hole
736/736T 	100	.00015 - 50	6 - 15,000 2 - 25,000	-50 to 120	15	stainless steel	10-32 tapped hole

TRIAxIAL ACCELEROMETERS

Triaxial units contain three mutually perpendicular accelerometers in a compact housing, allowing the evaluation of three dimensional motion.

	Sensitivity, mV/g	Acceleration Range, g	Frequency Response, Hz all channels ±10%	Temperature Range, °C	Weight, grams	Case Material	Mounting
733 	10	.00015 - 250	2 - 4,000	-50 to 120	23	hardcoated aluminum	10-32 captive screw
755G* 	3	.000005 - 1.5	6 - 4000	-50 to 80	1900	titanium	—
993A 	100	.0002 - 50	2 - 2,000	-50 to 120	90	hardcoated aluminum	¼ - 28 captive screw

* Ultra low noise sensor.

ACCELEROMETERS

Model HA8 Hand-Held Vibration Probe - Acceleration Output

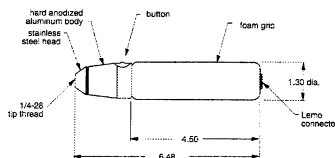


Features:

- Rugged
- Easy to use
- Shock protected with intrinsic overload protection
- Built-in electronics
- Miswiring protection

Applications:

- Predictive maintenance
- Bearing analysis
- Ship board machinery
- Heavy machinery
- Compressors



Model HA8

SPECIFICATIONS

DYNAMIC

Sensitivity, $\pm 5\%$, 25 °C	100	mV/g
Acceleration Range	50	g peak
Amplitude Nonlinearity	1	%
Frequency Response $\pm 10\%$:		
screwed down using mounting stud	2 - 2000	Hz
through probe tip	5 - 500	Hz

ELECTRICAL

Power Requirement, voltage source	18 - 30	VDC
current regulating diode	2 - 10	mA
Electrical Noise, equiv. g, nom.:		
Broadband, 2.5 Hz to 25 kHz	100	μg
Spectral	10 Hz	$\mu\text{g}/\sqrt{\text{Hz}}$
	100 Hz	$\mu\text{g}/\sqrt{\text{Hz}}$
	1000 Hz	$\mu\text{g}/\sqrt{\text{Hz}}$
Output Impedance, max.	100	Ω
Bias Output Voltage, nom.	10	VDC
Grounding	case isolated, internally shielded	

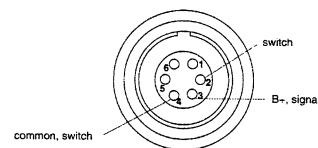
ENVIRONMENTAL

Vibration Limit	500	g peak
Shock Limit	100	g peak
Electromagnetic Sensitivity, equiv. g.		$\mu\text{g}/\text{gauss}$

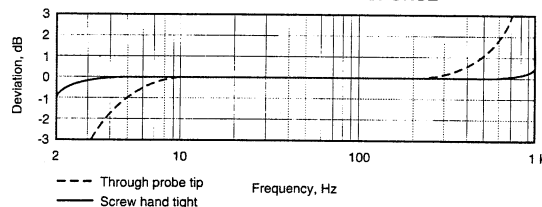
PHYSICAL

Weight	150	grams
Case Material	stainless steel head, hard anodized aluminum body	
Grip Material	black neoprene foam	
Output Connector (socket)	Lemo EGG1B306CNL	
Mating Connector (pins)	Lemo FGG1B306CNADA62 (R6LM)	

Connector Pinout



TYPICAL FREQUENCY RESPONSE



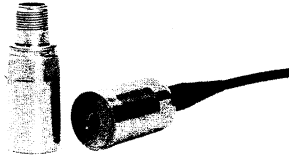
Due to continued research and product development, Wilcoxon Research reserves the right to amend this specification without notice.

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ACCELEROMETERS

Model 793L

Low Frequency Piezoelectric Accelerometer

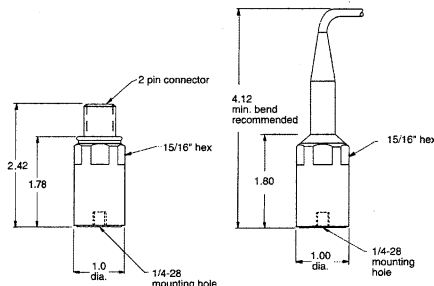


Features:

- Intrinsic safety certified option
- High sensitivity
- Ultra low-noise electronics for clear signals at very low vibration levels
- Low frequency capable
- Filtered to eliminate high frequencies
- Hermetic seal
- ESD protection
- Miswiring protection

Applications:

- Machinery monitoring
- Rotating machinery
- Low vibration level measurements
- Laboratory research



Model 793L

Model 793LS

SPECIFICATIONS

DYNAMIC

	793L	UNITS
Sensitivity, $\pm 5\%$, 25 °C	500	mV/g
Acceleration Range ¹	10	g peak
Amplitude Nonlinearity	1	%
Frequency Response, nom.:		
$\pm 5\%$	0.6 - 700	Hz
$\pm 10\%$	0.4 - 1000	Hz
$\pm 3\text{ dB}$	0.2 - 2300	Hz
Resonance Frequency, mounted, nom.	15	kHz
Transverse Sensitivity, max.	7	% of axial
Temperature Response	see graph	

ELECTRICAL

Power Requirement, voltage source	18 - 30	VDC
current regulating diode ^{1,2}	2 - 10	mA
Electrical Noise, equiv. g, nom.:		
Broadband, 2.5 Hz to 25 kHz	10	μg
Spectral		
2 Hz	1.5	$\mu\text{g}/\sqrt{\text{Hz}}$
10 Hz	0.4	$\mu\text{g}/\sqrt{\text{Hz}}$
100 Hz	0.2	$\mu\text{g}/\sqrt{\text{Hz}}$
Output Impedance, max.	100	Ω
Bias Output Voltage, nom.	10	VDC
Grounding	case isolated, internally shielded	

ENVIRONMENTAL

Temperature Range	-50 to 120	°C
Vibration Limit	250	g peak
Shock Limit	2500	g peak
Electromagnetic Sensitivity, equiv. g.	5	$\mu\text{g}/\text{gauss}$
Humidity Limit	100	% relative
Base Strain Sensitivity	0.001	$\text{g}/\mu\text{strain}$

PHYSICAL

Weight	135	grams
Case Material	stainless steel	
Mounting	1/4-28 tapped hole	
Output Connector	MIL-C-5015 2-pin	
Cabling ³ : Mating Connector	Amphenol 97-3106A-10SL-4S (R6)	
Recommended Cable	J5, coaxial, polyurethane jacket, 30 pF/ft	

- NOTES:** ¹ Signal distortion can occur when measuring high vibration levels, especially with long cables. A 24-30 VDC powering source is recommended for minimizing distortion.
² A maximum current of 6 mA is recommended for operating temperatures in excess of 100 °C.
³ Cabling for "S" option model is J61, coaxial, polyurethane jacket, 30 pF/ft.

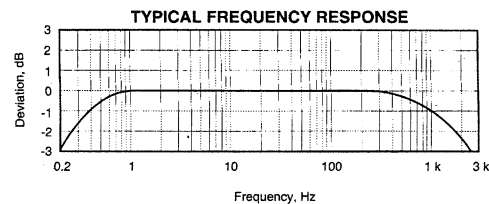
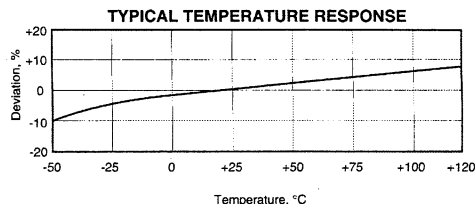
Connections

- Pin A Signal, power
Pin B Common

OPTIONS: Intrinsic safety "E" (793LE) certified by Factory Mutual for use in the following environments: Classes I, II, III, Div. 1, Groups A, B, C, D, E, F, G, 10 ft. sealed integral cable "S", temperature sensor "T", customer specified sensitivity, filtering, cable length.

ACCESSORIES SUPPLIED: SF6 mounting stud.

ACCESSORIES AVAILABLE: R62-J5-XX cable assembly, magnetic mounting bases, cementing studs, power supplies, amplifiers, signal conditioners.

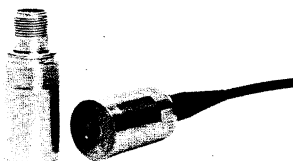


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ACCELEROMETERS

Model 793 Industrial Piezoelectric Accelerometer

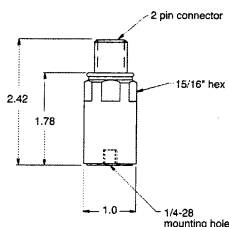


Features:

- Intrinsic safety certified option
- Rugged, general purpose
- Corrosion-resistant
- Priced economically for single and OEM quantities
- Hermetic seal
- Standardized sensitivity
- Ground isolated
- ESD protection
- Miswiring protection

Applications:

- Engine monitoring
- Machinery monitoring
- Ship board machinery
- Shock & vibration
- Predictive maintenance
- Shock pulse measurements



Model 793

SPECIFICATIONS	793	UNITS
DYNAMIC		
Sensitivity, $\pm 5\%$, 25°C	100	mV/g
Acceleration Range ¹	50	g peak
Amplitude Nonlinearity	1	%
Frequency Response:		
$\pm 5\%$	3 - 5,000	Hz
$\pm 10\%$	2 - 7,000	Hz
$\pm 3\text{ dB}$	1 - 15,000	Hz
Resonance Frequency, mounted, nom.	26	kHz
Transverse Sensitivity, max.	5	% of axial
Temperature Response	see graph	
ELECTRICAL		
Power Requirement:		
voltage source	18 - 30	VDC
current regulating diode ^{1,2}	2 - 10	mA
Electrical Noise, equiv. g, nom.:		
Broadband, 2.5 Hz to 25 kHz	700	μg
Spectral		
10 Hz	10	$\mu\text{g}/\sqrt{\text{Hz}}$
100 Hz	5	$\mu\text{g}/\sqrt{\text{Hz}}$
1000 Hz	5	$\mu\text{g}/\sqrt{\text{Hz}}$
Output Impedance, max.	100	Ω
Bias Output Voltage, nom.	12	VDC
Grounding	case isolated, internally shielded	
ENVIRONMENTAL		
Temperature Range	-50 to 120	°C
Vibration Limit	500	g peak
Shock Limit	5000	g peak
Electromagnetic Sensitivity, equiv. g.	15	$\mu\text{g}/\text{gauss}$
Humidity Limit	100	% relative
Base Strain Sensitivity	0.004	$\text{g}/\mu\text{strain}$
PHYSICAL		
Weight	110	grams
Case Material	stainless steel	
Mounting	1/4 - 28 tapped hole	
Output Connector	MIL-C-5015 2-pin	
Cabling: Mating connector	Amphenol 97-3106A-10SL-4S (R6)	
Recommended Cable	J5, coaxial, polyurethane jacket, 30 pF/ft	

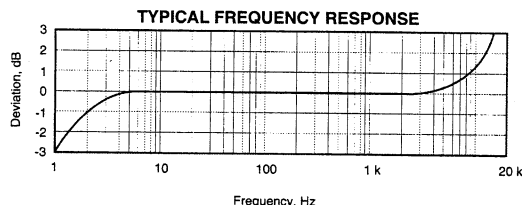
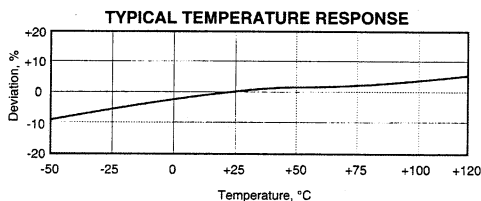
NOTES: ¹ Signal distortion can occur when measuring high vibration levels, especially with long cables. A 24-30 VDC powering source is recommended for minimizing distortion. Current regulating diode must be at least 2 mA plus 1 mA per 1000 pF of cable capacitance.

² A maximum current of 6 mA is recommended for operating temperatures in excess of 100 °C.

OPTIONS: Intrinsic safety "E" (793E) certified by Factory Mutual for use in the following environments: Classes I, II, III, Div. I, Groups A, B, C, D, E, F, G, 10 ft. sealed integral cable (see 793S/SG), temperature sensor "T", ultra low noise "U", customer specified filtering, sensitivity, cable length.

ACCESSORIES SUPPLIED: SF6 mounting stud.

ACCESSORIES AVAILABLE: Magnetic mounting bases, cementing studs, cable assembly R62-J5-XX, power supplies, amplifiers, signal conditioners.



Due to continued research and product development, Wilcoxon Research reserves the right to amend this specification without notice.

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